

Neurosplash

2024 한국뇌신경과학회 고등인지분과 하계 워크숍

일시: 2024. 8. 18(일) – 19 (월)

장소: 청풍리조트 컨벤션홀



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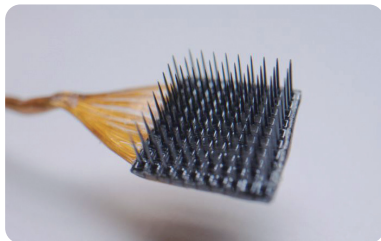
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2024 한국뇌신경과학회

고등인지분과 하계 워크숍

일시: 2024. 8. 18 (일) – 21(월)

장소: 청풍리조트 컨벤션홀

1 박 2 일로 개최되는 본 하계 워크숍에서는 한국뇌신경과학회의 정식 분과인 고등인지분과 회원들이 모여 연구실의 최신 연구 주제와 방향을 소개하고 논의하는 기회를 마련함으로써, 고등인지 신경과학 분야의 최신 트렌드를 이해하고 서로 유용한 정보를 공유하고자 합니다.

일자	시간	프로그램
8 월 18 일	13:00-14:00	Registration
	14:00-14:10	개회사
	14:10-14:40	포스터 홍보의 시간
	14:40-17:40	포스터 세션
	17:40-18:00	기념사진 촬영 및 체크인
	18:00-19:30	저녁 만찬
	19:30-20:30	토론 세션
8 월 19 일	10:00-10:30	분과 운영회의
	10:30-11:00	시상식 및 폐회사

Abstract List

A brain-wide investigation of the neural substrates of associative learning (<i>Seungyub Lee, Geun Ho Im, Seong-Gi Kim, HyungGoo Kim</i>)	1
Diversity of lateral habenula neurons in reward and aversive processing (<i>Jongwon Yun, Huynnggoo kim</i>)	2
Dopaminergic computation of positional and rotational proximity (<i>Rin Wang, Seungyub Lee, Jongwon Yun, Hyeon Jang, Hyunggoo Kim</i>) .	3
Temporal credit assignment in alcohol use disorder (<i>Jiwon Park, Manjae Kwan, Young-Chul Jung, Dongil Chung</i>)	4
Differential Reactivation of Task-related Firing Patterns of Place Cells be- tween the Subiculum and CA1 in a Visual Scene Memory Task (<i>Jae- Min Seol, Su-Min Lee, and Inah Lee</i>)	5
Neuropsychosocial Measures Linked to Problematic Gaming Cross-Sectionally and Longitudinally in Developing Youth (<i>Suyeon Song, Marc Potenza, Woo-Young Ahn</i>)	6
Processing temporal information during enjoyable experiences (<i>Hee-Dong Yoon, Hyeon-Ae Jeon</i>)	7
Contextual Information Enhances Attentive States to Mitigate Temporal Bias in Time Estimation (<i>Jungtak Park, Hyeon-Ae Jeon</i>)	8
Sequential Rule Generalization through Reinforcement Learning (<i>Taehyun Yoo, Hyeon-Ae Jeon</i>)	9
Neural mechanisms underlying value normalization adaptive to changes in choice-relevant information (<i>MinHo Hwang, Dongil Chung</i>)	10
The cortical contribution to individual difference in emotional valence pro- cessing (<i>Jaewoong Hwang, ChiHye Chung</i>)	11
The Implicit Dynamics of Hierarchical Sequence Learning (<i>Minho Shin, Hyeon-Ae Jeon</i>)	12
Neural basis establishing the competition-driven social hierarchy (<i>Minsung Sim, Hyogyun kim, ChiHye Chung</i>)	13
Encoding of task event boundaries by subicular neurons in virtual environ- ments (<i>Su-Min Lee, Inah Lee</i>)	14
Inactivation of the intermediate hippocampus differentially affects the pre- limbic and infralimbic cortex virtual context working memory (<i>Hee- Seung Ha, Eun-Hye Park, Inah Lee</i>)	15
Neural Correlates of Risk and Reward Signals in Smokers Using an MRI- Compatible Vaping Device: A Preliminary Study (<i>Eunhwi Lee, Jeung- Hyun Lee, Joshua W. Brown, Woo-Young Ahn</i>)	16

Differential Contributions of Human Hippocampal Subregions to Object-in-Context Memory at Different Learning Stages (<i>Sorin Jeong, Ji Yeon Kim, Jae-Min Seol, Ji Sun Kim, Sang Ah Lee, Inah Lee</i>)	17
Coordinated Computations for Naturalistic Memory Encoding and Retrieval in Hippocampal Neural Subspaces (<i>Dasom Kwon, Jungwoo Kim, Seng Bum Michael Yoo, Won Mok Shim</i>)	18
Decision-making for others' time and money (<i>Sunmin Kim, Dongil Chung</i>)	19
Neural Processes Linking Interoception to Moral Preferences Aligned with Group Consensus (<i>JuYoung Kim, Minyoung Kim, Daon Lee, Hackjin Kim</i>)	20
The sex specific impact of metabolomic reprogramming in depression (<i>Jimin Nam, Seungjae Zhang, Seulgi Kang, Jeongmi Lee* , and ChiHye Chung*</i>)	21
Effects of uncontrollable stress on an association of a conditioned stimulus with the absence of the unconditioned stimulus (<i>Nam-Heon Kim, Su-Yeon Sim, Jung-soo Han</i>)	22
Effect of blocking entorhinal cortical dopamine receptors on value-dependent conditioned responses (<i>Jaehun Kim, Minyeong Seo, Min Whan Jung</i>)	23
Neural mechanisms of allocentric spatial representation using visual landmarks in the hippocampus (<i>Eunho Lee, Inah Lee</i>)	24
Threat-dependent defensive behavior mediated by CGRP neurons in parabrachial nucleus (<i>Buyng Min Chung, Gyeong Hee Pyeon, June-Seek Choi, Yong Sang Jo</i>)	25
Consistency of EEG Microstate among Various Contexts (<i>Jiyeon Jeong, Jayoon Choi, Minjae Jo, Min-Suk Kang</i>)	26
Distractor control modulates theta-band power (<i>Hansol Lee, Sun-Young Park, Hyungjin Hwang, Min-Suk Kang</i>)	27
Balance of innovation and imitation for sociocultural evolution (<i>Seungdae Baek, Hyeonsu Lee, Minsoo Kang, Se-Bum Paik</i>)	28
Stable receptive field in the early visual pathway for domain-general object recognition (<i>Minjun Kang, Seungdae Baek, Se-Bum Paik</i>)	29
Emergence of acoustic functional maps from neuronal interference in auditory cortex (<i>Jeonghwan Cheon, Se-Bum Paik</i>)	30
Heroes and villains: Opposing narrative roles engage neural synchronization in the inferior frontal gyrus (<i>Hayoung Ryu, Minue Kim</i>)	31
Amygdala Response in Koreans to Surprised Facial Expressions (<i>Seohyeon Lee, Minue Kim</i>)	32
Neural coding hierarchy of summary statistics and categories across mouse primary visual and posterior parietal cortices (<i>Young-Beom Lee, Gaeun Jung, Oliver James, Doyun Lee, Yee-Joon Kim</i>)	33
Dynamic representation of spatial and non-spatial information by single units in the primate hippocampus during contextual object discrimination (<i>Seukhwan Shin, Jaemin Seol, Bona Lee, Choong-Hee Lee, Sumin Lee, Heung-Yeol Lim, Yuji Naya, Joonyeol Lee, Inah Lee</i>) . . .	34
Disentangling Pain Factors: Neural Insights via Variational Autoencoder (<i>Jihoon Han, Sungwoo Lee, Choong-Wan Woo</i>)	35

Negativity bias in the retrieval of context-emotion associations in depression (<i>Minsok Koo, Sang Ah Lee</i>)	36
Separate effects of emotional valence and semantic informativeness on emotional episodic memory (<i>Hye Bin Yoo, Seongyoon Yang, Sang Ah Lee</i>)	37
Neural dynamics of context transitions in spontaneous thoughts (<i>Kwon Heo, Dasom Kwon, Won Mok Shim</i>)	38
Do Social Preferences Depend on Context? Children's Probabilistic Social Decision-Making Based on Affiliation and Dominance (<i>Gakyung Kim, Jeongyeon Shin, Woo-Young Ahn, Sang Ah Lee</i>)	39
Investigation of context-dependent decision making along the olfactory pathway (<i>Jongrok Do, Doyun Lee</i>)	40
Hippocampal Correlates of Individual Differences in Temporal Episodic Memory in Aging (<i>Maria Jieun Hwang, Sang-Eon Park, Sang Ah Lee</i>)	41
Value and spatial signal processing in the rodent granular retrosplenial cortex (<i>Daeul Kong, Eun Ju Shin, Min Whan Jung</i>)	42
Neural Mechanisms Linking Temporal and Directional Predictions in Sensory-Motor Behaviors (<i>Sanghum Woo, Sua Bae, Jeongwon Kim, Geonwoo Kim, Yechan Hwa, Joonyeol Lee</i>)	43
The Double-Edged Sword of Event Segmentation in Children's Memory (<i>Yu Jin Rah, Sang Ah Lee</i>)	44
Neurobehavioral impacts of a social observer on risky decision-making in cigarette smokers (<i>Hee Young Seon, Cheolin Yoo, Dongil Chung</i>) . . .	45
Enhanced Performance of Predictive Models in Within and Between-Person Pain using Multi-echo fMRI (<i>Suhwan Gim*, Eui-Jin Jung*, Dong Hee Lee, Jae-Joong Lee, Choong-Wan Woo, †</i>)	46
Dynamic Pain Regulation: From Onset to Recovery (<i>Yuseon Park, Jungwoo Kim, Eunjin Lee</i>)	47
Neural Mechanisms of Music-Induced Analgesic Effect through Personally Pleasant Music (<i>Eui-Jin Jung*, Elise Desbarats*, Mathieu Roy, †, Choong-Wan Woo, †</i>)	48
Context-dependent influence of CRH neurons in the paraventricular nucleus of the hypothalamus on active avoidance behavior (<i>Hayeon Lee, Han Eul, June-Seek Choi, Yong Sang Jo</i>)	49
Neural Signatures of Individual Differences in the Influence of Interoceptive Signals on Preference Decisions (<i>Yuri Kim, Jinhee Kim, Hackjin Kim</i>)	50
Effects of Ketamine treatment on the uncontrollable stress-induced memory impairments (<i>Sang-Pil Park, Jung-Soo Han</i>)	51
Neural representations in MPFC and insula encode individual differences in estimating others' preferences (<i>Hyeran Kang, Kun Il Kim, Hackjin Kim</i>)	52
Effect of physical effort on dopamine activities in the ventral and dorsal striatums (<i>JiHwan Kim, Jaebin Yoon, HyungGoo R. Kim</i>)	53
A derivative-like dopaminergic computation across valences (<i>Mijoo Park, Jongwon Yun, Hogyu Choi, Hyunggoo Kim</i>)	54

From vision to action during closed-loop naturalistic interaction: Spatially specific reward signals in visual areas and their behavioral impact (<i>Royoung Kim, Jiwoong Park, Kendrick Kay, Won Mok Shim</i>)	55
Value representation in striatum-projecting PPC neurons (<i>Sangwoo Kim, Jiwon Yoon, Eun Ju Shin, Min Whan Jung, Seung-Hee Lee*</i>)	56
The role of CGRP neurons in suppressing addictive behavior (<i>Jihyun Lee, Gyeong Hee Pyeon, June-Seek Choi, Yong Sang Jo</i>)	57
Modulation of inequity aversion through social hierarchy in mouse model (<i>Yujeong Jeong, Sukwoon Song, Jihyun Noh, Jeongyeon Kim</i>)	58
Shared memory representations in real-world social interactions (<i>Jiun Choi*, Jiwoong Park*, Won Mok Shim, †</i>)	59
Dysfunction of the primary visual cortex leads to visual impairments in a Grin2b-mutant mouse model of autism (<i>Jong-Yeon Yoo, Ilsong Choi, Eunjoon Kim, Seung-Hee Lee</i>)	60
Suppression of Risky Foraging Behavior Requires Increased Central Amygdala Activity at Discrete Time Points (<i>Jaeyong Lee, June-Seek Choi</i>)	61
Weakening of cortical connectivity and multisensory integration in the Shank3 Δ 14–16 mice (<i>Seung-Mi Oh, You-Hyang Song, Hae-Yong Park, Jun-Ho Song, Eunjoon Kim, Se-Bum Paik, Seung-Hee Lee*</i>)	62
Plcg1 dependent neural plasticity is required for dynamic hippocampal CA2 network processing social memory in mice (<i>Sukwoon Song, Dongwoo Nam, Jeongyeon Kim</i>)	63
Chemogenetic activation of the lateral habenula facilitates the association of CS with the absence of the US in Pavlovian appetitive conditioning (<i>In-Beom Jin, Jung-Soo Han</i>)	64
Egocentric spatial information decoding by population activity in the mouse retrosplenial cortex (<i>Daehyun Oh, Yoonsoo Yeo, and Jeehyun Kwag</i>)	65
Roles of retrosplenial parvalbumin-positive interneurons in the neural representation of environmental geometry (<i>Jiyeon Yang, Jeehyun Kwag</i>)	66
Impaired Egocentric Neural Encoding of Environmental Geometry in the Retrosplenial Cortex in the 5XFAD Mouse Model of Alzheimer’s Disease (<i>Yoonsoo Yeo, Jeehyun Kwag</i>)	67

A brain-wide investigation of the neural substrates of associative learning

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Previous studies on the neural mechanisms of associative learning in animals often focused on specific brain regions. Here, we investigated the mechanisms of cue-reward associative learning in a brain-wide manner using functional MRI (fMRI) and dopaminergic (DA) manipulations. Mice ($n=7$) underwent two fMRI sessions under anesthesia. On day one, mice received a visual cue (5Hz bilateral flicker), followed by pairing the visual cue with optogenetic stimulation (20Hz) on DA neurons in the ventral tegmental area (VTA). Finally, an identical visual cue probed the neural correlates of association. On day two, we conducted the same experiment, except for the second session containing only optogenetic stimulation as a control. As expected, the flickering stimuli activated the visual cortex and superior colliculus, and VTA stimulation activated the striatum. Surprisingly, the visual-optogenetic pairing additionally activated the medial prefrontal (mPFC), septum, and cingulate cortex ($p<0.05$), underscoring their potential role in associating cues and rewards. What is the functional role of prefrontal activation? Recent studies suggest that mPFC DA is involved in salience processing. We hypothesized that salient stimuli activate the mPFC and facilitate associative learning. To test this, we trained head-fixed mice in two groups for a visual cue-reward association task, with habituation (8 days) in darkness followed by an association task (8-11 days): Group 1 ($n=12$) underwent habituation without reward, leading to high salience on the visual cue and reward in the later association task. Group 2 ($n=6$) received random rewards during habituation to mitigate the salience of the reward. Interestingly, associative learning was significantly faster in Group 1 ($p<0.05$), highlighting the importance of reward salience in facilitating associative learning. We plan to investigate mPFC DA and spiking activity's roles in associative learning through optogenetic and electrophysiological methods.

Keywords: associative learning, fMRI, mPFC, salience

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Diversity of lateral habenula neurons in reward and aversive processing

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Lateral habenula (LHb) is implicated in aversive learning, depression, as well as reward processing. From the perspective of reinforcement learning, dopamine (DA) activity encodes the temporal difference (TD) error, a specific form of reward prediction error (RPE). Electrophysiological recordings in nonhuman primates show that spiking activity in LHb is inversely related to dopamine activity, and there is an inhibitory anatomical connection from LHb to DA neurons, suggesting that LHb activity may be a major contributor to the DA RPE signal.

Our recent study demonstrated that DA activity is a moment-by-moment temporal derivative of the value function. Using a similar experimental paradigm, we set out to test that LHb neurons would show inverse TD error in a moment-by-moment manner. We trained mice in virtual reality for simple association tasks. In track 1, mice moved passively toward a goal location where they obtained a drop of water as a reward. In track 2, mice received an electrical tail shock (0.4mA, 150ms duration) at the end of the track. We added teleportation conditions where mice suddenly moved backward to a different position and continued moving on. If an LHb neuron encodes inverse TD RPE, it would show transient inhibition in response to backward teleport in the shock track, whereas it would show transient excitation in response to backward teleport in the reward track.

We recorded 246 neurons in LHb using Neuropixels 1.0 from 5 mice. To our surprise, we observed a small subset of neurons show activity patterns consistent with the inverse TD error prediction. LHb neuronal activity shows striking diversity. Only about 27.7% of LHb neurons are excited by unexpected shock and inhibited by unexpected reward. Furthermore, the responses to teleportations in the 76.6% of LHb neurons are not consistent with the predictions from the inverse TD RPE prediction. Our results suggest that LHb populations in mice are more diverse than previously proposed.

Keywords: lateral habenula, reward, aversion, virtual reality

Acknowledgements: This work was supported by 1711196916 and IBS-R015-D1.

Dopaminergic computation of positional and rotational proximity

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Living with multiple potential goals, we take into account our current status and reward statistics when making decisions. Here, we developed a novel task to model this more realistic reward environment in a controlled laboratory setting. In a square chamber, an LED cue is positioned at the center of each side, with a water port installed beneath each cue. When one of the four visual cues is randomly activated for a second, a drop of water is dispensed from the port below the activated cue. Trained mice could recognize the cue and approach the port to obtain water. We observed a decrease in the rate of false licks (licks to ports without water) and latency to the correct port. The mouse's precise head position was analyzed frame-by-frame using DeepLabCut. We found that the latency increased with the mouse's distance to the reward (13 sessions from 3 mice). When the cue was potentially out of the mouse's visual field (> 120 deg from the midline), lick latency was significantly longer. Interestingly, when mice could see the cue, latency significantly decreased with the visual angle of the cue. To monitor dopamine (DA) activity, we injected a dopamine sensor (DA2m) into the ventral striatum (VS). We found that the DA response to the cue significantly decreases with distance. Additionally, DA response to cue was negatively correlated with the angle of the cue (13 sessions from 3 mice). These results suggest that mice predict the time to reach the reward taking into account the distance and relative angle of their body. Then the time-to-reward is used to discount dopaminergic cue responses in the VS, demonstrating highly adaptive computation in the naturalistic environment.

Our freely moving experimental setup enables us to further test how social interactions influence decision-making. By adding more animals, we aim to investigate changes in behavioral strategies and responses, giving insights into the decision-making process under social and competitive situations.

Keywords: ventral striatum, decision-making, spatial relationship

Acknowledgements: This work was supported by IBS-R015-D1.

Temporal credit assignment in alcohol use disorder

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[Introduction] The Credit Assignment Problem (CAP) refers to the challenge of associating actions with their temporally delayed consequences. Given temporal uncertainty, solving CAP is crucial for making decisions that have long-term consequences such as in matters related to health. However, it is still unclear if there is a specific association between individual's substance use and their comprehension of causal relationships between action and outcomes. In the current study, we investigated whether and how the mechanisms of CAP-solving are altered in individuals with alcohol use disorder (AUD). [Methods] 29 individuals with AUD and 31 healthy controls (HC) were recruited. Participants were asked to make a series of choices between two options where the reward contingency associated with each option could only be learned through experience. Most importantly, reward outcomes were temporally delayed, such that individuals were informed about the consequence of each choice only after an uncertain delay. A hidden structure was composed of the temporal delay of one trial most predominantly, followed by the other cases (e.g., a delay of zero or two trials). After each choice, the reward outcomes delayed to the corresponding trial were summed and displayed on the screen. [Results] We found that only in HC, the proportion of choosing the better option increased in the second half of the learning task. Against alternative models, we found that individuals' learning behavior was best explained by a mixture model between a model-based learning strategy—attributing credit for observed outcomes to their past choices based on their beliefs about the task's temporal structure (1-BACK or 2-BACK association)—, and a model-free learning strategy—attributing credit to their most recent choice when updating their expectations about future states ('future state values' in the temporal difference (TD) learning). Notably, we found that only in HC, the extent to which individuals assigned credit to the immediately preceded choice (1-BACK) increased as they experienced the learning structure. This indicates that HC, but not AUD, successfully learned the most common causal relationship between actions and outcomes under temporal uncertainty. [Conclusion] The current study showed that excessive alcohol use is related to impaired comprehension of causal relationships between actions and outcomes under temporal uncertainty.

Keywords: Addiction, Learning, Credit assignment

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Differential Reactivation of Task-related Firing Patterns of Place Cells between the Subiculum and CA1 in a Visual Scene Memory Task

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Spatial and non-spatial representations are both essential for episodic memory. It has been demonstrated that spatial representations of place cells are reactivated during sharp-wave ripples (SWRs). However, whether non-spatial, task-related information (e.g., scene and choice) is also reactivated in SWRs is unclear. We examined this by reanalyzing the data used in our previous study (2022) in which rats were trained to perform a visual scene-based spatial choice task while single units were recorded from CA1 and subiculum (SUB). As reported previously, individual place fields were defined by spiking phases in relation to theta rhythm. We analyzed a group of place cells co-activated during single SWR events. In the SWRs recorded from SUB and CA1, we found ‘task-related reactivations’, specific reactivation patterns in which cells with similar selective firing for scene or choice were reactivated together. Among SWRs with significant reactivations, task-related reactivation occurred more in SUB than in CA1. We also found a unique type of task-related reactivation that represented scene and choice information simultaneously, which was more prevalent in SUB than in CA1. Next, we compared the participation rates in SWR events between different cell groups during task-related reactivations. SUB cells with multiple theta-based fields (MF cells) showed a higher SWR participation rate than single-field (SF) cells, while CA1 MF cells did not. This may be attributed to MF cells that had more selective task-related representations than SF cells in SUB. Our study shows that non-spatial representations of place cells are reactivated during SWRs as robustly as spatial representations, and SUB may play key roles in amplifying and conveying such task demands. Specifically, in SUB, the MF cells’ activities may underlie the higher proportion of task-related reactivations in SWRs and the multiplexing capability for qualitatively heterogeneous information.

Keywords: Hippocampus, Subiculum, Sharp-wave ripple

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Neuropsychosocial Measures Linked to Problematic Gaming Cross-Sectionally and Longitudinally in Developing Youth

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The elevated prevalence of Internet Gaming Disorder among adolescents highlights the critical need to explore both the risk and protective factors associated with gaming addiction during this developmental stage. The dual systems model suggests that an imbalance between the development of the prefrontal cortex and the limbic regions heightens adolescents' sensation-seeking drives while their self-regulatory capabilities are still maturing, thus increasing their vulnerability to risky behaviors. A disadvantaged social environment can further intensify these tendencies. Existing theoretical frameworks, including the I-PACE model primarily focus on neuropsychological constructs of IGD, such as impaired response inhibition and altered reward anticipation. However, they often overlook the social contexts in which adolescents engage. This study proposes an integrated neuropsychosocial framework to examine both the risk and protective factors influencing gaming addiction among youth. Utilizing a dataset from the ABCD study, we applied multivariate regression analyses with elastic net regularization and cross-validation to investigate neuropsychosocial factors of gaming addiction among approximately 3,000 adolescents, both cross-sectionally and longitudinally. Our findings indicate that higher trait impulsivity, negative familial and peer relationships, school disengagement, and extensive multiplayer gaming time are significant predictors of both current and future gaming addiction risk. Conversely, positive school involvement, alongside neural alterations in the inferior frontal gyrus and rostral anterior cingulate cortex during response inhibition tasks, and hyperactivation in the medial orbitofrontal cortex during reward anticipation, emerged as protective factors against gaming addiction. This study empirically tested theoretical models of IGD and highlights the necessity of integrating social-environmental factors to fully understand gaming addiction among adolescents.

Keywords: Adolescent, Gaming addiction, Neuropsychosocial

Acknowledgements:

Processing temporal information during enjoyable experiences

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Our estimation of elapsed time often deviates from veridical time, which is a phenomenon referred to as time distortion. While various factors such as attention, arousal, and memory are recognized contributors to time distortion, the impact of fun on subjective time distortion has not been thoroughly investigated. Hence, our study aims to scrutinize the relationship between subjective fun and time distortion, along with the associated cognitive mechanisms.

In our experiment, participants viewed recordings of soccer games that varied in levels of fun. To gauge their subjective experience of fun, we asked the participants to press a button whenever they experienced a sense of enjoyment. For each video, participants reported fun ratings on a scale from 1 to 100 and provided estimates of the video's duration. Using an eye-tracker, we measured pupil size as an indicator of cognitive resource allocation during fun-filled moments.

For the fan group, time distortion was positively correlated with subjective fun, and they exhibited a larger average pupil size during moments of fun compared to non-fun moments. More specifically, when the number of button presses increased, duration estimates also increased. These results imply that the number of moments characterized by the participants' subjective fun led to duration overestimation. This aligns with numerous experimental findings indicating that the more a stimulus is segmented into multiple scenes or events, the greater the overestimation of its duration. Interestingly, these propensities were not apparent in the control group. We suggest that soccer fans, who have more knowledge and experience with soccer, are more susceptible to the details contained in the stimuli compared to the controls. This difference may have led soccer fans to pay more attention to the contents of the video, leading them to think that fun videos lasted longer than boring videos. Overall, the degree of fun directly impacts duration estimation.

Keywords: Subjective time estimation, Time distortion, Subjective fun

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Contextual Information Enhances Attentive States to Mitigate Temporal Bias in Time Estimation

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The ability to accurately estimate the passage of time is essential for many aspects of human life. Previous research has demonstrated that context can influence time estimation. However, these studies have overlooked how an awareness of context affects time estimation. This study investigates whether individuals can improve their performance in time estimation tasks by utilizing contextual information. Participants were divided into two groups: an informed group provided with contextual information and an uninformed group without contextual information. We assessed the impact of contextual information on errors in duration estimates using a Bayesian generalized linear model. The results indicated that the informed group made fewer errors in their time estimations. Further analysis using an iterative Bayesian model revealed that the informed group paid more attention to time than the uninformed group. Additionally, functional magnetic resonance imaging showed increased activation in the attention network of the informed group during the time estimation task. Overall, these findings suggest that contextual information enhances attentive states, thereby mitigating temporal bias in time estimation.

Keywords: Time estimation, Contextual information, Iterative Bayesian model

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Sequential Rule Generalization through Reinforcement Learning

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The ability to generalize rules in novel environments is a remarkable characteristic of humans. While previous studies have explored rule generalization based on perceptual similarity between stimuli, little is known about how humans generalize sequential knowledge. This study aims to investigate the generalization of rules through the abstraction of sequences. We devised a sequence learning task where participants judged the correctness of relational structures among visual stimuli, guided by given cues and feedback. We hypothesized that participants would generalize new cues by leveraging their associations with known cues to infer sequential rules. Participants were divided into two groups: an experimental group who pre-trained on rules associated with some cues, and a control group without prior rule knowledge. To analyze the experimental group's behaviors, we developed four reinforcement learning (RL) models, each characterized by distinct state-spaces corresponding to different generalization strategies. Specifically, the first model included states that correctly mapped cues to sequential rules. The second and third models depended solely on cue shape and cue color, respectively. The fourth model mapped cues to sequential rules in a reversed manner. A mixture-of-experts model was then employed to elucidate how participants utilized these strategies to judge sequential rules, by estimating the reliabilities of the strategies. Our finding revealed that the experimental group more rapidly formed correct sequential rules for new cues compared to the control group. The RL model incorporating the correct generalization strategy provided a superior fit for the experimental group's behavior. Additionally, in the mixture-of-experts model, the reliability assigned to the correct generalization strategy increased as the experiment progressed. These findings provide insights into the computational mechanisms underlying the generalization of sequential rules.

Keywords: Generalization, Sequence, Reinforcement learning

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Neural mechanisms underlying value normalization adaptive to changes in choice-relevant information

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Choices are influenced by context changes. For instance, changes in an item's availability (e.g., sold-out) may affect purchasing decisions. Previous studies suggested that value normalization, a theory regarding the normalized neural representation reflecting the full range of available information, accounts for the impact of an inferior option—choice-irrelevant information—on three-alternative forced choices. However, it remains unknown whether and how changes in choice-relevant information affect individuals' choices. We propose a Two-step value normalization model (TSVN) that offers a potential explanation for the impacts of changes in the available options, using a context-adaptive value normalization process. To test our hypothesis, we used an EEG recording alongside a novel three-option gambling task. During the task, on some trials, one option became unavailable ('Sold-out' trials) before the choice cue, prompting individuals to make decisions from the remaining two options. Against other normalization models, our proposed TSVN model, which assumes the second-step normalization to be applied between the normalized values of the two available options, best explained individuals' choice patterns across all trials. Consistent with the model's interpretation regarding the involvement of choice-irrelevant information, we found that the normalized value of the sold-out option was associated with the single-trial EEG amplitude in the P3 channel at 410-540 ms and in the FT9 channel at 460-500 ms after the Sold-out cue. Furthermore, these neural-level sensitivities were consistent with model-estimated individual differences, such that the neural instantiation of the sold-out option at the P3, followed by the FT9, was associated with the extent to which individuals normalize values at the second step. These results provide a neuromechanistic account of how changes in choice-relevant information during choice valuation affect individuals' decision processes.

Keywords: Value normalization; Decision-making; EEG; Computational modeling

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The cortical contribution to individual difference in emotional valence processing

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Human emotions are complex and diverse. Evaluating the emotional value of environments or objects is important for social interaction and survival. This evaluation can vary among individuals, especially those who experience chronic stress or suffer from depression. Despite its significance, it's currently unknown which specific brain regions and mechanisms are involved in this process. In this study, our aim is to investigate individual differences in emotional valuation and the effects of stress using animal models, with a focus on the activity of the prefrontal cortex. We set up an experiment with two different environments: one where animals were exposed to positive stimuli (chocolate) and another where they received negative stimuli (foot shocks). In contextual preference tests, we found significant individual variations in the preference for the environment associated with both positive and negative stimuli. We used fiber photometry with the calcium indicator protein GCaMP8f and found a significant correlation between decreased activity in the prefrontal cortex and a stronger preference for the stimulated context. Additionally, when we used the acute learned helplessness (aLH) model, we observed a similar negative correlation between prefrontal cortex activity and preference for the stimulated context. Overall, these findings suggest that activity in the prefrontal cortex may play a crucial role in emotionally evaluating stimuli, potentially influencing individual preferences for environments associated with positive and negative experiences.

Keywords: emotional valence, mPFC, individual difference

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The Implicit Dynamics of Hierarchical Sequence Learning

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Humans construct their world knowledge hierarchically to behave more flexibly and efficiently. Yet, little is known about how we implicitly construct internal representations of the environment hierarchically from a stream of information without explicit feedback. To address this issue, we designed a sequence learning task where participants passively responded to a continuous stream of visual stimuli. Specifically, 50 participants were instructed to respond to the keys corresponding to the stimulus position during fMRI scanning. The stimulus presentation was governed by a hierarchical structure that participants were unaware of. This hierarchical structure consisted of a few hidden states, where a particular stimulus was predominantly presented, or all stimuli were randomly presented. We hypothesized that participants would implicitly learn this hidden structure, leading to faster response times.

To examine human behaviors, we introduced several ideal observer models. The first, a hierarchical observer model, optimally learns the hidden hierarchical structure using a hidden Markov model. As an alternative, a Dirichlet-categorical model predicts the next stimulus based on previous first-order transitions between stimuli. As this model lacks any information on hidden states, it was termed as the flat observer model. To compare the performance of each model with that of humans, we computed surprise, uncertainty, and belief update based on the models' predictions. The results showed that human behaviors were best explained by the hierarchical observer model. Post-experiment surveys confirmed that the participants did not have explicit knowledge of the task's hierarchical structure.

These results suggest that people can construct their world knowledge hierarchically, even without awareness. Additionally, using fMRI, we identified the neural mechanisms related to the implicit learning of sequences with a hierarchical structure.

Keywords: Implicit hierarchical sequence learning, Ideal observer model, fMRI

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Neural basis establishing the competition-driven social hierarchy

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Social animals, including humans, naturally establish hierarchies for resource optimization and cooperation. The social hierarchy formation involves direct competition or indirect comparison based on sensory information. To determine which social factors are more important in establishing social hierarchy, we employed the social hierarchy test with indirect competition, the urine marking test (UMT), before and after the tests with direct competition, the tube test (TT). The social rank was unstable before the direct competition, however after the competition, social ranks became consistent with significant correlations between TT, highlighting the importance of direct competition in establishing social hierarchy. We also found that the activity of the medial prefrontal cortex (mPFC) was correlated with rank; the higher the social rank, the higher the mPFC activity. Considering the distinct activation patterns of the mPFC and the involvement of dopamine receptors, we examined whether dopamine signaling in the mPFC contributes to shaping the social hierarchy. We found that blocking either D1- or D2-type receptors in the mPFC during the TT disrupted the social hierarchy. Additionally, the social rank-dependent differences of mEPSC amplitudes were abolished after dopamine receptor blockade. Our findings suggest that direct competition may modulate dopaminergic signaling in the mPFC, which is essential in establishing social hierarchy.

Keywords: Social hierarchy, Prefrontal cortex, Dopamine

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Encoding of task event boundaries by subicular neurons in virtual environments

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How is hippocampal information utilized for goal-directed navigation? To answer this question, the subiculum is the cornerstone due to its anatomical position situated at the juncture between the hippocampus and other cortical areas. We hypothesize that the subiculum represents a space according to task demands. To test this, three male Long-Evans rats were trained to navigate on a circular track in a virtual reality (VR) task in which event boundaries change flexibly. Specifically, the starting position for each trial varied pseudo-randomly among three locations (90°, 180°, 270°), while rewards were given at a fixed point (0°). Two distinct VR environments were used, so the rat had to choose one of two water ports at the reward point depending on the given environment to receive a reward. The trial ended when the rat circled back to the starting point. During the learning phase, we recorded single-unit spiking activities and local field potentials in the subiculum and hippocampus simultaneously using a 24-tetrode multi-electrode drive. Our preliminary results revealed that some subicular neurons encoded task events such as trial initiation, which were crucial for delineating the task structure. The neurons flexibly adjusted their firing fields based on the changing distances between events, dictated by the varying starting positions. Notably, such event coding was present even before the rats learned contextual association of behavioral responses. Moreover, the neurons exhibited context-dependent rate modulation within their event-specific firing fields. After the rat learned the task, subicular neurons dynamically represented the boundaries of task events according to the VR contexts and starting locations. Combining these observations with prior reports of spatial boundary coding in the subiculum, we propose a generalized function of the subiculum in episodic memory that represents behaviorally significant boundaries in space and events.

Keywords: Hippocampus, Subiculum, Episodic memory

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Inactivation of the intermediate hippocampus differentially affects the prelimbic and infralimbic cortex virtual context working memory

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It is widely known that the hippocampus (HP) and medial prefrontal cortex (mPFC) are crucial to working memory (WM), but their interaction is not fully understood. Specifically, the infralimbic cortex (IL) of the mPFC receives heavier projections from the intermediate HP (iHP) than the prelimbic cortex (PL) does. Here, we explored the neural dynamics between the IL/PL and iHP in male Long-Evans rats (n=5) running on a cylindrical treadmill during a WM task in virtual reality (VR). The task consisted of three phases: sample, delay, and test. First, rats experienced one of four VR contexts (Forest, City, Room, Playground) during the sample phase, and then underwent a 3-sec delay in a gray tunnel. In the following test phase, in which the rats made either a left or right lick as a response, and, if correct, honey water was provided as reward (120 trials/session). We implanted a hyperdrive in PL and IL for single-unit recording (24 tetrodes) with or without injecting muscimol (MUS, 0.3-0.5 μ L/site) into the iHP. MUS injections into the iHP decreased WM accuracy (86% to 62%, $p < 0.001$). Preliminary analysis showed that mPFC neural activity was influenced by VR context and choice. Neurons in PL (n=49) and IL (n=39) were similarly modulated by the VR context and choice during the sample and delay phases. In the sample phase, iHP inactivation altered PL's context-driven activity ($p < 0.01$), while IL exhibited reduced neural pattern modulation by VR context ($p < 0.05$) and choice ($p < 0.01$). Notably, choice-related neural activity decreased in IL during the delay ($p < 0.001$). 9.5% of IL cells showing significant modulation by task-related variables represented choice responses during both the sample and delay phases, but not in PL. Meanwhile, 13.3% of PL neurons encoded context and choice responses across phases, not in IL. Our findings suggest that the PL represents task-relevant information and the IL organizes the information according to the temporal structure of the task.

Keywords: medial prefrontal cortex, intermediate hippocampus, working memory

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Neural Correlates of Risk and Reward Signals in Smokers Using an MRI-Compatible Vaping Device: A Preliminary Study

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Investigating the neural mechanisms of reward and risk processing is crucial for understanding impaired decision-making in addiction, where the evaluation of reward and risk is often distorted. While it's crucial to utilize primary rewards to capture drug-specific neural processes, less is known about reward and risk encoding in the context of actual drug use. This study aims to fill this gap by investigating how reward and risk processing differ between smokers and non-smokers, using both monetary and nicotine rewards in a structured gambling task and a novel MRI-compatible vaping device. We employed fMRI to examine the neural correlates of reward prediction errors and risk prediction errors. Preliminary fMRI results with monetary rewards indicated significant activation in the striatum correlated with reward prediction errors. Additional findings highlighted risk prediction error-related activations in the posterior parietal cortex (PPC) and the dorsolateral prefrontal cortex (dlPFC). Notably, our results show that smokers exhibited distinct neural responses from non-smokers, suggesting their altered reward and risk encoding. Specifically, the attenuated response of the striatum supports the hypothesis of addiction-related reward system desensitization. Furthermore, the increase in PPC activation during risk prediction error processing in smokers may suggest altered risk encoding in nicotine addiction, necessitating further investigation to unravel the interplay between risk processing and addictive behaviors. These preliminary findings using monetary rewards set the groundwork for forthcoming experiments with the novel MRI-compatible vaping device to directly assess neural responses to nicotine rewards. The use of a direct nicotine reward would further allow for a more accurate depiction of the addiction-specific neural processes. It is our expectation that this study will shed light on our understanding of the distinct neural processing of primary drug rewards.

Keywords: Reward and risk, Nicotine, fMRI

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Differential Contributions of Human Hippocampal Subregions to Object-in-Context Memory at Different Learning Stages

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The hippocampus associates information about stable aspects of the environment ('context') and objects to form object-in-context (OiC) memory. In human studies, little is known about the exact roles of the hippocampus during OiC memory acquisition compared to retrieval. In the current study, we investigated the contributions of hippocampal subregions to the OiC learning process in an Object-Context Association Task (O-CAT). In this task, thirty participants were required to associate 4 visually similar objects with 2 VR environments by trial-and-errors. Specifically, participants navigated a U-shaped linear track in one of two environments per lap. In each lap, the four objects were presented one at a time in a random sequence, and they pressed 'YES' or 'NO' button to indicate whether the object is associated with the current context. Before and after the acquisition phase, an object without any background was presented to test OiC memory retrieval. The task was conducted inside an MR scanner. Behavioral results revealed that most participants learned the O-CAT within the first half of the task. fMRI pattern similarity (PS) analysis in the hippocampus during retrieval showed increased PS between objects associated to the same context in the CA1 region. Interestingly, during the acquisition, the CA1 maintained high PS only for objects sharing the same context ($p < 0.05$ and $p < 0.01$ for the 1st and 2nd half trials, respectively), while the CA23DG region showed a decreased PS for objects associated with all contexts ($p < 0.05$ for the 1st vs. 2nd half trials). These preliminary results suggest that the OiC memory representations are highly adaptive, resulting from the differential roles of the subregions. The CA23DG may orthogonalize OiC memory representations, helping to reduce interference from comparable information, whereas the CA1 can group OiC memory representations with a common spatial context.

Keywords: Hippocampus, Object-in-Context Memory, fMRI

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Coordinated Computations for Naturalistic Memory Encoding and Retrieval in Hippocampal Neural Subspaces

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Our naturalistic experiences are organized into memories through multiple processes, including novelty encoding, memory formation, and retrieval. However, the neural mechanisms coordinating these processes remain elusive. Using fMRI data acquired during movie viewing and subsequent narrative recall, we examined hippocampal neural subspaces associated with distinct memory processes and characterized their relationships. We quantified novelty in character co-occurrences and the valence of relationships and estimated event memorability. Within the hippocampus, the novelty subspaces encoding each type exhibited partial overlap, and these overlapping novelty subspaces aligned with the subspace involved in memorability. Notably, following event boundaries, hippocampal states within these subspaces aligned inversely along a shared coding axis, predicting subsequent recall performance. This novelty-memorability alignment was selectively observed during encoding but not during retrieval. Our findings offer novel insights into how the hippocampus dynamically coordinates computations underlying memory encoding and retrieval at the population level to transform ongoing experiences into enduring memories.

Keywords: Hippocampus, aligned subspace, coordinated computation, memory, novelty encoding, naturalistic stimuli

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Decision-making for others' time and money

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People often make choices on behalf of others, such as decisions related to investments or scheduling. However, it remains unknown how individuals decide for others without knowing others' preferences. Here, we used a lottery game in which participants made choices between a safe lottery (a guaranteed payoff) and a risky lottery (a 25, 50, or 75% chance of receiving a higher payoff or no payoff) and functional near-infrared spectroscopy (fNIRS) to investigate how individuals make gamble choices for both themselves and anonymous others. Participants made choices regarding time (waiting time) and money (bonus) lotteries for themselves and a randomly chosen partner, respectively in both gain- and loss-only frames (i.e., 'Self-Time', 'Partner-Time', 'Self-Money', 'Partner-Money' conditions). Participants showed a longer response time (RT) to make decisions for themselves compared with decisions for their partner, and a longer RT when deciding between lotteries with monetary payoffs than that with time payoffs. At the presentation of the cue indicating for whom they are making the choice, individuals who showed a larger RT difference between decisions for others and that for themselves showed a higher HbO signal in the frontopolar region on the decisions for others compared with the choices for themselves, and vice versa. Estimated behavioral characteristics revealed that individuals were in general more loss-averse when deciding about time payoffs than monetary payoffs, and that they had higher value sensitivity (i.e., making more consistent choices) in Partner-Money than in Partner-Time condition. Individuals' empathy score was positively correlated with their value sensitivities, estimated from the decisions made for their partner. Together, our results suggest that social context affects individuals' decisions under risk in different domains.

Keywords: Self-Other decision-making, Empathy, fNIRS

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Neural Processes Linking Interoception to Moral Preferences Aligned with Group Consensus

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Aligning one's decisions with the prevailing norms and expectations of those around us constitutes a fundamental facet of moral decision-making. When faced with conflicting moral values, one adaptive approach is to rely on intuitive moral preference. While there has been theoretical speculation about the connection between moral preference and an individual's awareness of introspective interoceptive signals, it has not been empirically examined. This study examines the relationships between individuals' preferences in moral dilemmas and interoception, measured with self-report, heartbeat detection task, and resting-state fMRI. Two independent experiments demonstrate that participants' interoceptive awareness and accuracy are associated with their moral preferences aligned with group consensus. In addition, the fractional occupancies of the brain states involving the ventromedial prefrontal cortex and the precuneus during rest mediate the link between interoceptive awareness and the degree of moral preferences aligned to group consensus. These findings provide empirical evidence of the neural mechanism underlying the link between interoception and moral preferences aligned with group consensus.

Keywords: Moral preference, Resting state, Dynamic connectivity

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The sex specific impact of metabolomic reprogramming in depression

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It is widely recognized that there is a higher prevalence of depression in females compared to males; however, the underlying mechanisms involving metabolites remain largely unknown. Recently, the sub-chronic variable stress (SCVS) protocol has been proposed to mimic female vulnerability to stress. We previously employed this model to perform a non-targeted metabolomics comparison and found higher levels of arachidonic acid (AA) in the plasma of stressed females compared to female controls, whereas AA levels decreased in the plasma of stressed males. Therefore, we investigated the physiological and behavioral impact of AA-targeted metabolomic reprogramming in the SCVS model. Reprogramming of AA levels was achieved by inhibiting major enzymes required for AA generation: fatty acid amide hydrolase (inhibited by URB597) and phospholipase A2 (inhibited by quinacrine). Intriguingly, URB597 administration during the stress, but not quinacrine, successfully ameliorated depressive behaviors in females. Moreover, stress-induced enhancement of spontaneous excitatory postsynaptic currents in the basolateral amygdala (BLA) was also successfully restored to control levels upon URB597 administration. Our observations suggest that metabolic differences may mediate the sex differences in depression, and proper metabolic reprogramming could serve as a powerful female-specific therapeutic approach for depressive disorders.

Keywords: Sex differences of depression, Basolateral amygdala, Metabolite reprogramming

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Effects of uncontrollable stress on an association of a conditioned stimulus with the absence of the unconditioned stimulus

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Associating a conditioned stimulus (CS) with the absence of an unconditioned stimulus (US) generates conditioned inhibition. Conditioned inhibition, along with conditioned excitation, is a fundamental behavioral mechanism for guiding adaptive behavior. We have demonstrated that the lateral habenula (LHb) is a critical neural structure mediating conditioned inhibition. It has also been reported that stressful treatments activate the LHb and that the LHb is hyperactivated in patients with depression. Therefore, we examined the effects of uncontrollable stress on conditioned inhibition. Animals received 60-min restraint + 60 tail shocks and, 24 hr later, explicitly unpaired training of a light cue and food, one of the procedures generating conditioned inhibition. The conditioned inhibitory property of the light was assessed in a retardation test; an acquisition of excitatory learning is delayed in the paired training of the light and food followed. Unstressed animals showed a delay, whereas the stressed animals did not show a delay. These results suggest that stress elicits LHb dysfunction, leading to conditioned inhibition impairments.

Keywords: Conditioned inhibition, Stress, Lateral habenula

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Effect of blocking entorhinal cortical dopamine receptors on value-dependent conditioned responses

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A growing body of evidence indicates the involvement of the hippocampus in value-based decision making. Our research further revealed that neurons in layer II of the medial entorhinal cortex (MEC) robustly encode signals related to reward value. To explore the role of dopamine in MEC value processing, we examined dopamine receptor distribution in the MEC and the consequences of receptor blockade on value-dependent conditioned responses. Dopamine D1 and D2 receptors were found across layers I, II, and V, with minimal presence in layer III. Blocking dopamine receptors in the MEC disrupted value-dependent anticipatory licking behavior in probabilistic classical conditioning, suggesting the involvement of the dopaminergic system in MEC value processing. It remains to be determined how dopamine modulates value-related MEC neural processes.

Keywords: Dopamine, Entorhinal cortex, Value

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Neural mechanisms of allocentric spatial representation using visual landmarks in the hippocampus

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It has been known that the hippocampus uses allocentric visual landmarks to represent the animal's current location. The neural mechanisms underlying such spatial computation, however, are largely unknown. We have shown previously that place cells in the CA1 subregion of the hippocampus produce place fields robustly with a small number of visual landmarks as rats navigate in a virtual environment (Shin et al., 2022). However, more landmarks needed to be provided to activate the place cells in CA3 in the same rats. Our findings led us to hypothesize that the CA3 network requires more 'scene-like' visual stimuli to represent a location allocentrically in the VR environment compared to CA1. To test this hypothesis, we parametrically manipulated the number of visual landmarks in the VR environment to see at which point in that stimulus continuum the CA3 network started to exhibit spatial coding. Specifically, we recorded single units from CA1 and CA3 as body-fixed male Long-Evans rats (n=8) ran along a virtual track to obtain water rewards (30 μ L) at random locations. Upon arriving at the end of the track (i.e., lap), the rats were teleported back to the start position, unknowingly to the animal. After training, recordings began in an environment where only the track and visual landmarks were present. Only two landmarks were present initially on both sides of the track. Then, after finishing every 15 laps, the number of landmarks increased by two, up to 20. We expect that the increased number of landmarks will trigger an immediate corresponding increase in the number of place fields in CA1. In contrast, CA3 is expected to abruptly increase the number of place fields only in response to a high density of landmarks, a so-called scene-like stimulus. Furthermore, CA1 place fields are expected to be sensitive to the introduction of additional landmarks, altering their firing patterns, whereas CA3 is likely to maintain a stable representation once place fields have emerged.

Keywords: Hippocampus, place cell, visual landmark

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Threat-dependent defensive behavior mediated by CGRP neurons in parabrachial nucleus

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Recent studies suggest that neurons expressing calcitonin gene-related peptide (CGRP) in the parabrachial nucleus (PBN) convey aversive information from diverse sensory modalities and signal a general alarm to the forebrain. All of the studies commonly found that CGRP stimulation increases freezing behavior among various defensive responses. However, it remains unclear whether these neurons can also induce other types of defensive behavior besides freezing. To investigate this, we selectively activated or inactivated CGRP neurons during two different unconditioned stimuli (US) that carry distinct features: a chasing robot (70 cm/s for 3 s), which mimics a predator-like threat where threat imminence is perceptible, and an electric footshock (0.3 mA for 1 s), which is invisible to the animals, making the distance to the threat indeterminable. Our results demonstrated that CGRP stimulation paired with the chasing robot increased fleeing responses, while the same CGRP stimulation resulted in increased freezing responses rather than fleeing when paired with the footshock, compared to their respective control groups. Moreover, optogenetic inactivation of CGRP neurons during both types of US led to a significant reduction in fear responses, resulting in decreased freezing and fleeing behaviors. We then further examined whether CGRP neurons also encode the intensity of threats by increasing the speed of the chasing robot (90 cm/s). We compared the fleeing behavior of animals exposed to the high speed to those exposed to the previous speed paired with CGRP stimulation and found that animals exhibited similar levels of fleeing responses in both conditions. Taken together, our findings suggest that CGRP neurons in the PBN function as a core alarm system, enhancing the representation of threat information by reflecting threat features and amplifying the perceived intensity of the threat.

Keywords: CGRP, Fleeing, Freezing

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Consistency of EEG Microstate among Various Contexts

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We build a mental model of the world based on their statistical regularities and neural responses during the resting state should reflect such if we ruminate previous learning sessions. EEG microstate analysis recorded during resting interval has shown that the states and their dynamics reflect several cognitive variables and their progression (Bréchet et al., 2019). Here, we seek to identify a building of mental model by obtaining EEG microstates. Participants performed three types of decision-making tasks (exploration, decision-making, & navigation) for reward, and each task session consists of approximately 5 min task period and 5 min post-task rest period. In addition, while those tasks look different, they share a common structure despite the fact that participants should apply different rules depending on context to maximize the reward. Three microstates were computed from the rest period before task (baseline rest), task period, and the post-task rest period and they were similar. In addition, those microstates are similar across participants, suggesting that they reflect basic cognitive functions. However, the transition pattern of the microstates varies. In particular, the transition dynamics between the task period and the post-task rest period are distinguished, suggesting that transition pattern of microstates reflects sequential context

Keywords: fMRI-EEG, Microstate, Learning

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Distractor control modulates theta-band power

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Resistance against external distractions is a hallmark ability of working memory. We can enhance this ability if we know when and what type of distractors occur ahead. Here, we seek to identify what brain mechanisms are responsible for controlling distraction in relation to accumulated knowledge of the distractors. According to the gate theory of working memory, our brain has gating mechanisms that learn to control distraction such that the brain opens the gate to the targets but closes to the distractors. We hypothesized that theta-band oscillation should reflect the gate operation based on that frontal midline theta (FM θ) carries several cognitive control variables. Participants performed a change-detection task in which distractors were presented during the retention interval in 50% of trials while their electroencephalogram (EEG) was recorded. We found that frontal midline theta (FM θ) power decreased after a presence of targets and maintained. However, a presence of distractors elevated it. This elevation of FM θ power following distraction reduced with task progress.

Keywords: working memory, distractor control, frontal midline theta (FM θ)

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Balance of innovation and imitation for sociocultural evolution

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Sociocultural evolution, the process by which cultural behavior changes over time within populations, has been documented in both human and nonhuman animals (Whiten, 2007; Clutton-Brock, 2021). Innovators, who introduce novel behaviors not previously seen in their populations, are believed to drive this evolution by exploring new ecological niches, diversifying actions and traits, and ultimately enhancing overall fitness (Tebich, 2016). However, it is often observed that imitators, who replicate the actions of others, tend to dominate various social groups (Reader, 2016). This raises questions about the conditions and mechanisms by which innovators and imitators can induce social evolution. Using a computational approach with a multi-agent system, we examined how innovators and imitators influence population behavior evolution. We conducted simulations to model agent survival, with each agent possessing a trait that affects their energy efficiency. Initially, we tested whether innovators alone could drive social evolution and found that their random trait variations did not significantly improve lifespans, as these traits did not reach optimal survival values. Interestingly, the introduction of imitators, who randomly copied others' traits, led to increased lifespans and a convergence toward optimal traits. This indicates that the presence of both innovators and imitators is crucial for social evolution. Moreover, adjusting the ratio of innovators to imitators showed that the evolutionary effect was maximized with a few innovators and many imitators. Our results suggest that the combination of a minority of innovators and a majority of imitators is essential for population evolution, shedding light on the mechanisms of population evolution and the importance of the balance between innovation and imitation.

Keywords: Random imitation, Social learning, Multi-agent simulation

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Stable receptive field in the early visual pathway for domain-general object recognition

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Recognizing objects under dynamic environments is an essential ability for animals (Kudithipudi et al., 2022), but it is a challenging task for conventional deep neural networks (DNNs). When DNNs are trained under variations of image domains, they often lose previous information, which is called catastrophic forgetting (McCloskey & Cohen, 1989). However, the mechanisms producing such distinction between the two systems remain elusive. Unlike DNNs' early layers, which start from random noises and are trained to datasets, the brain's early visual pathway inherently has Gabor-like receptive fields (Ringach, 2002) that remain stable throughout a lifetime (Godecke et al., 1997). Here, we propose that inherent Gabor filters in the early layer, motivated by the receptive fields of cells in the primary visual cortex (V1), enable robust object recognition under dynamic domain changes. To test this, we modeled receptive fields of V1 as Gabor filters and incorporated them into the first layer of DNNs (Krizhevsky et al., 2012) to model biological visual systems. Then, we trained networks sequentially on two image domains, Photo and Sketch images, known for their different low-level features, such as spatial frequency. We first confirmed that conventional DNNs could not classify Photo images when trained on the Sketch images, whereas those with fixed Gabor filters mostly maintained their performance. This effect is generalized to other domains, underscoring the generalizable effect of stable Gabor filters. Next, to probe the underlying mechanism of our model, we analyzed representations within latent space from the internal layers. Our model produced consistent object manifolds encompassing diverse domains, which may improve the networks' ability to recognize objects robustly. In conclusion, Gabor filters in early layers act as significant structures for domain-general object recognition, emphasizing the critical role of the brain's stable early visual pathway.

Keywords: Object recognition, Deep neural network, Primary visual cortex

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Emergence of acoustic functional maps from neuronal interference in auditory cortex

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Pitch perception, the ability to perceive the periodicity of a harmonic sound, is observed in higher animals and is considered a fundamental function of auditory cognition. Through cochlear mechanical filtering, harmonic sounds are spectrally decomposed into frequency components that are integer multiples of the fundamental frequency. The brain remarkably extracts sound periodicity from this decomposed tonotopic representation. Previous research reported that the brain employs a spatially graduated tuning map of periodicity in the auditory pathway for pitch extraction (Langner, 2009; Baumann, 2011; Barton, 2012). However, the developmental origins of these acoustic functional maps in the ascending auditory pathway remain unclear. Here, we propose that a periodicity map can spontaneously develop from neuronal interference between ipsilateral and contralateral tonotopic maps. We assumed that bilateral tonotopic maps, when converging with a tilt, create wiring with varying characteristic frequencies. We demonstrate that wiring of distinct characteristic frequencies encodes both frequency and periodicity information via beat interference. We modeled tonotopic and periodotopic maps using carrier and beat frequencies as interference results. The resultant periodotopic map displayed a wide range of periodicities, as seen in the auditory cortex. We also verified that the tonotopic gradient remains intact, which could elucidate the emergence of a periodotopic map in a tonotopically organized ascending pathway. To further support our hypothesis, we compared our model maps with optically recorded data from the primary auditory cortex (A1) in cats (Langner, 2009). Our model and A1 maps exhibited significant two-dimensional correlations in tonotopy and periodotopy. Overall, our model offers a plausible developmental mechanism for the formation of a periodotopic map through neuronal interference in the auditory pathway.

Keywords: Pitch perception, Periodotopic map, Auditory cortex

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Heroes and villains: Opposing narrative roles engage neural synchronization in the inferior frontal gyrus

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Using movie-watching fMRI, we sought to identify the neural underpinnings of processing opposing narrative roles (i.e., protagonists and antagonists) while the participants watched a film with highly engaging, socially rich audiovisual narratives. We utilized two publicly available fMRI datasets in which participants watched parts of *Sherlock* and *The Grand Budapest Hotel*. We annotated and concatenated the TRs during which the protagonists (*Sherlock* and *Gustave*) and antagonists (*Mycroft* and *Dmitri*) appeared on screen for each character. Then, neural synchronization was calculated using inter-subject correlation (ISC) by randomly sampling TRs to match the TR length of the protagonist or antagonist who had less screen time. Paired t-tests were performed to identify parcels whose ISC differed between the protagonists vs. antagonists. We repeated this procedure for 1,000 iterations and found parcels that consistently showed significant differences in at least 95% of the iterations (i.e., ≥ 950 iterations) in both datasets for generalizability. Across both datasets, the left inferior frontal gyrus (IFG) and left orbitofrontal cortex (OFC), showed significantly higher ISC (Bonferroni-corrected $p < .0005$) during the appearance of the protagonists. Conversely, the auditory cortex exhibited significantly higher ISC when antagonists appeared on screen. We found that the IFG and OFC are conditionally involved in processing opposing narrative roles. A possible interpretation is that information integration and emotional empathy are employed depending on a character's narrative role during movie watching. Our findings add to the current literature that different narrative roles involve high-order social cognitive processing and may set the stage for a better understanding of person perception in more naturalistic settings.

Keywords: IFG, Social perception, Movie-watching fMRI

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Amygdala Response in Koreans to Surprised Facial Expressions

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Although surprise is often considered an ambiguously valenced emotion, there are surprised facial expressions that most people evaluate as either positive or negative. Previous research conducted with Americans revealed that amygdala activation is greater when viewing negative surprised facial expressions compared to positive ones. However, in the present fMRI study involving Koreans, the opposite result was found. This was also the case for surprised facial expressions with a mask, even though the emotional valence of the negative and positive stimuli was similar. Hence, it can be hypothesized that, unlike Americans, amygdala activation in Koreans is influenced by factors other than emotional valence.

Keywords: Amygdala, Surprised facial expressions, fMRI

Acknowledgements:

Neural coding hierarchy of summary statistics and categories across mouse primary visual and posterior parietal cortices

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Despite growing evidence that visual systems exploit the intrinsic correlational structure of the world to represent summary statistics and scene categories, it remains unclear how the brain processes concrete sensory information into increasingly abstract information from mean feature values to visual categories. We characterized the population-level neural coding of summary statistics and categories from calcium signals imaged in primary visual cortex (V1) and posterior parietal cortex (PPC) while head-fixed mice passively viewed or classified eight mean motion directions of randomly moving dots into two categories. Multivariate pattern analyses revealed the variance-dependent mean motion direction representation in V1 and the task-dependent representations of the mean and the category of motion direction in PPC. Neural representations of mean motion directions in V1 were systematically biased toward the center of the appropriate category during motion categorization task. This, together with the variance-discriminating pattern observed only for extremely heterogeneous stimuli, suggests that mouse V1 is well suited for global mean motion processing. Our finding of redundant encoding of summary statistical visual features across V1 and PPC suggest that summary statistical representations accumulate across the visual hierarchy, which may allow PPC to bind multiple levels of summary statistical representations into abstract task-oriented category signals.

Keywords: Summary statistical representation, Motion direction, Microendoscope imaging, Primary visual cortex, Posterior parietal cortex

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Dynamic representation of spatial and non-spatial information by single units in the primate hippocampus during contextual object discrimination

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It has been hypothesized that neurons in the hippocampus represent objects and the environmental context in which they appear for episodic memory. However, the types of neural correlates underlying object-in-context (OiC) memory has never been rigorously tested in nonhuman primates (NHPs). We used VR technology to characterize the neural firing patterns in the hippocampus while NHPs performed a OiC memory task in the VR environment. We recorded single units in the hippocampus while NHPs (n=2) performed the task in which animals navigated along a linear track in one of two VR environments using a joystick. During navigation, they were stopped at one of 8 different trial locations and encountered a pair of objects (chosen among six objects). NHPs were required to choose one of the objects that was rewarded in the current environment to obtain a water reward (0.2-0.4ml). Both NHPs successfully learned the task (>96% accuracy). Among the cells analyzed (n=143), some hippocampal units exhibited positional correlates; 51% (n=74) of single units were presumed spatial units with a spatial information score (bit/spike) of 0.25 or higher in at least one of the environments. We observed object-related units (n=33, 23%) that significantly increased their firing rates when specific objects were presented within a specific context. Interestingly, 45% of cells showing object-related firing exhibited spatial correlates, indicating conjunctive coding of spatial and non-spatial variables by single units. We also observed a shift between spatial and non-spatial event-related firing patterns according to the environment in some conjunctive neurons. These results suggest that spatial and non-spatial variables are multiplexed in the firing patterns of a large population of single neurons in the hippocampus of NHPs in a contextual manner, highlighting the complexity of hippocampal coding and its potential role in memory formation.

Keywords: Non-human primate, hippocampus, object in context recognition

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Disentangling Pain Factors: Neural Insights via Variational Autoencoder

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Pain involves complex bottom-up stimulus-driven mechanisms and top-down cognitive, affective, and personalized processes. Our research uses functional Magnetic Resonance Imaging (fMRI) data to identify factors contributing to pain perception. We applied Variational Autoencoders (VAEs), a non-linear dimensionality reduction method, to create a latent space that enhances individual separation while interpreting latent factors like stimulus intensity through generative models. In this latent space, individuals form distinct clusters, preserving the representation of pain stimulus intensity at both group and individual levels. These findings were consistent across both training and independent test datasets. Further analysis identified representative vectors for each factor. A meta-analysis of the generated brain data from these vectors, using the VAE decoder, revealed that subtracting the representative vector shifted results from pain-related keywords to those associated with memory, social, and default mode networks. This demonstrates that we can manipulate and interpret the VAE latent space to effectively disentangle pain processing factors. Our findings suggest that individualized representations reveal unique patterns of pain processing in the brain, extending beyond group-level representations. Specifically, VAEs can separate group-level, subject-specific, and stimulus-intensity factors, offering a nuanced understanding of pain processes. We propose that these factors can be interpreted as external and internal elements of pain processing. External factors might include experimental design aspects (e.g., stimulus intensity), while internal factors could encompass autobiographical memories and mental states, which can be further divided into group-level and individual-level components. These separable factors can significantly enhance the precision of pain assessments and interventions, emphasizing the value of personalized approaches in pain management.

Keywords: Personalized Pain Processing, Variational Autoencoders (VAEs), Functional Magnetic Resonance Imaging (fMRI)

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Negativity bias in the retrieval of context-emotion associations in depression

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In our daily lives, we often experience multiple emotionally significant events in the same place or context. Evidence suggests that emotions are bound to these spatial contexts in memory and are retrieved when we revisit these locations. Disproportionate recall of negative over positive emotions in such contextual memories could indicate a person's susceptibility to affective disorders such as depression. 23 healthy participants (mean age: 28.9) performed a scene-emotion association task and completed the Beck Depression Inventory-II to measure their depression-like symptoms. During the encoding part of the task, a neutral background scene was presented, followed by two separate emotional foreground scenarios (Negative, Neutral, or Positive) on top of the same scene in subsequent order. During retrieval, previously shown neutral background scenes, without the foreground emotional images, were presented, and participants were asked to rate the emotional valence associated with each scene. We observed significant differences between emotion conditions in both encoding and retrieval, confirming that emotions were well associated with the background. Interestingly, while valence ratings during encoding were not affected by depression, associated valence ratings during retrieval showed a correlation with depression. Specifically, in the mixed condition, valence ratings exhibited a wide range of individual differences. Those with higher levels of depression tended to evaluate backgrounds associated with mixed emotions more negatively. This suggests that people's emotional biases toward episodic memory do not emerge at the moment of experiencing the event but rather begin during the process of retrieving the memory.

Keywords: Emotion, Depression

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Separate effects of emotional valence and semantic informativeness on emotional episodic memory

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Emotion and semantics both influence episodic memory. We aimed to understand their concurrent and separate effects on episodic memory representations using a mixed emotional memory paradigm recorded in 3T fMRI. Participants encoded emotional experiences with two sequential foreground images (cues) presented on a shared background image (context), then later recalled the valence associated with the background. Sequentially paired cues were neutral-neutral, negative-negative, positive-positive, negative-positive, or positive-negative. Pairs were also categorized by their level of semantic information quantified by their verbal descriptions (lower vs. higher). We measured representational pattern similarity (PS) in medial temporal lobe for 1) the first vs. second emotional cues, and 2) background contexts during encoding vs. retrieval. Emotional cue PS differed more when emotional valence was shifted ($F(3, 31) = 11.12$, $p < 0.001$) regardless of semantic information level ($F(1, 33) = 1.16$, $p = 0.29$). The most prominent difference was for the positive-to-negative shift in emotion ($t = 2.54$, uncorrected $p < 0.008$), and this may be due to the salience of negative cues marking a change in experience. Context PS was higher for the same emotional valence compared to different valences ($F(1, 33) = 23.26$, $p < 0.001$). Unlike the neutral emotion encoding-retrieval PS that was consistently significant ($t = 2.47$, $p < 0.01$), PS for negative-negative ($t = 2.01$, $p = 0.03$) and negative-positive ($t = 3.00$, $p = 0.003$) pairs were significant only for cues with lower semantic information and for positive-positive pairs with higher information ($t = 2.40$, $p = 0.01$). In summary, the representation of emotion was influenced only by emotional valence, while the representation of associated context was affected by both valence and semantics. This suggests two separate yet interactive processes underlying emotional episodic memory.

Keywords: Emotional episodic memory, Visual associative memory, Medial temporal lobe

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Neural dynamics of context transitions in spontaneous thoughts

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Continuous experiences of external input can be organized into discrete events, and their boundaries are known to elicit neural responses in the default mode network (DMN). However, the neural mechanisms underlying transitions in internally generated spontaneous thoughts in the absence of external input remain unclear. This study examined the neural dynamics of internal context transitions during free speaking by quantifying topic changes in speech using a large language model (LLM). During fMRI scans, participants engaged in three speaking tasks: 1) freely expressing spontaneous thoughts (think-aloud), 2) alternately discussing two pre-given topics (topic-alternating), and 3) recalling the content of movies they had previously watched (movie-recall). We employed an LLM to obtain the next sentence prediction (NSP) score for each sentence of the participants' speech, which indicates the likelihood that one sentence follows another in a temporally and logically coherent manner. In the think-aloud and topic-alternating tasks, the proportion of sentences including participant-annotated topic boundaries decreased as NSP values increased, indicating that NSP measures derived from an LLM successfully capture transition moments in continuous thoughts. Crucially, in the para-cingulate network (PCN), we observed significant negative correlations between its linear regression coefficients at the end of each sentence and the NSP scores, even after controlling for semantic similarities and the pause length between sentences, revealing increased PCN responses as the degree of context changes increased. Moreover, neural state transitions in the DMN occurred more frequently after sentences with low NSP scores, suggesting that discontinuities in DMN states are associated with thought transitions. Our findings highlight the functional roles of the PCN and DMN in discretizing internally generated thoughts as mental context changes, possibly by regulating introspective processes.

Keywords: fMRI, Spontaneous thought, Speech

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Do Social Preferences Depend on Context? Children's Probabilistic Social Decision-Making Based on Affiliation and Dominance

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From early development, our social preferences are often biased toward particular social traits, such as affiliation or dominance. In real life, however, these traits are not static; they are probabilistic and vary by context and prior experiences. Moreover, they may be perceived differently, depending on whether the interaction involves oneself. How do we form social preferences from probabilistic interactions with multiple social partners? Using a computerized social decision-making task, we investigated how children (n=89, 5 to 9 years old) select a social partner for their “self” character to interact with when given a choice between two randomly presented characters among three, who yielded affiliative or dominant social outcomes with differing probabilities. In the affiliation context, we found that children strongly favored the character with a higher probability of displaying positive interactive behavior. In contrast, children's preferences in the dominance context varied: despite their ability to identify the most dominant figure in post-test questions, children showed high individual differences in their choice between the more or less powerful characters. This suggests an individual preference bias in social interaction, particularly involving one's own dominance within the social context. To further investigate how children learn and update the inferred value from others' observable social traits, we applied reinforcement learning models. These computational models allowed us to analyze how children adapted their social preferences based on the probabilistic outcomes, such as learning rate and loss aversion for affiliative or dominant outcomes. This approach effectively captured the context-dependent aspects of children's decision-making in response to varying social values. Our study underscores the dynamic nature of social learning from an early age, showing how children form preferences and make interaction choices in dynamic social environments.

Keywords: Social Cognitive Development, Social Decision-Making, Probabilistic Learning

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Investigation of context-dependent decision making along the olfactory pathway

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Contrary to the long-held belief that the prefrontal cortex (PFC) is critical for working memory and decision-making, recent findings suggest a negligible engagement of this area, especially in the context of olfactory delayed match-to-sample (DMS) tasks. This study reevaluates the neural substrates for working memory and decision-making by extending the focus beyond the PFC to the entire olfactory pathway. Specifically, we explore the possibility that early sensory areas might play a pivotal role in context-dependent sensory processing, directly influencing behavioral outcomes. Furthermore, this study addresses the challenge of disentangling match/nonmatch signals from choice (action selection) signals in neural responses. Traditional task paradigms have often blurred these distinctions, leaving ambiguity in interpreting whether neural activities represent perceptual decision (e.g., match/non-match) or choice (e.g., go/no-go). To address these questions, our approach includes a comprehensive survey of neuronal activity across various brain regions using advanced recording techniques (Neuropixels) to encompass a wide array of areas efficiently. Our findings reveal that in well-trained mice, early sensory areas such as the Anterior Olfactory Nucleus (AON) show significant involvement in memory retention and decision-making processes, contrasting with the minimal representation observed in traditionally acknowledged PFC sub-regions. Notably, the correlation between working memory representation in upstream areas like the AON and CA1 with behavioral performance challenges the conventional view of higher-order cortical areas as the primary sites of working memory and decision-making. Our results suggest a paradigm shift, advocating for a new perspective where sensory areas are integrally involved in sensory encoding, working memory, match/non-match, and choice, thus challenging the traditional hierarchical model of neural processing in perceptual decision making.

Keywords: Neuropixels, Decision-making

Acknowledgements:

Hippocampal Correlates of Individual Differences in Temporal Episodic Memory in Aging

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As we get older, episodic memory tends to decline due to a decrease in the function and volume of the medial temporal lobe (MTL). Research has shown that the ability to remember the sequential order of events is especially susceptible to deterioration not only in Alzheimer's disease but also in normal aging. In younger adults, individual temporal order memory performance is correlated with the modulation of hippocampal activity depending on the temporal distance (TD) between two events during recall. This study aims to determine whether a reduction in TD-modulated activity in the MTL is an indicator of age-related decline in temporal memory. Participants ranging from 50 to 86 years of age (n=120) performed a scene-based episodic memory task testing what, where, and when memory in the fMRI scanner. Temporal memory accuracy was lower than the other episodic memory components in older participants (what-when: $p < 0.001$; where-when: $p < 0.001$, and what-where: $p = 0.748$), and the correlation between TD and accuracy gradually declined in aging ($r = -0.42$, $p < 0.001$). In comparing between good and bad performers (median split), the good group showed a clear TD-modulated pattern in hippocampal activation, while the bad group did not. As in younger adults, older individuals with weak hippocampal TD-modulation showed bad temporal order memory ($r = 0.19$, $p < 0.05$). These results imply that hippocampal TD coding underlies our episodic retrieval of temporal order and, given its fragility, can be an informative marker of hippocampal decline in aging.

Keywords: Aging, Episodic Memory, fMRI

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Value and spatial signal processing in the rodent granular retrosplenial cortex

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The retrosplenial cortex (RSC) is intricately connected with diverse cortical and subcortical brain structures and is implicated in various cognitive functions such as spatial navigation, episodic memory, and value-based decision making. In rodents, the RSC comprises agranular and granular divisions, with the latter reciprocally connected with the hippocampus. In an effort to probe hippocampal-RSC interactions in value-based decision making, we examined the effects of inactivation and neural activity in the granular RSC (gRSC) in mice performing a dynamic foraging task in a figure 8-shaped maze. Bilateral inactivation of the gRSC using muscimol severely disrupted the animal's choice behavior in the task. Preliminary analysis employing reinforcement learning models suggests that gRSC inactivation increases choice bias while decreasing learning rate. Additionally, gRSC neurons recorded with one photon calcium imaging conveyed a diverse array of signals related to value-based decision making. Notably, they conveyed strong decision-value and chosen-value signals before and after the animal's selection of target, respectively, alongside encoding spatial information. A preliminary analysis indicated a positive relationship between decision-value signals and spatial information in neurons significantly responsive to decision value. These results suggest that the gRSC may conjunctively integrate value and spatial information to guide value-based decision making in freely navigating mice. Our future work aims to explore the underlying neural mechanisms by manipulating hippocampal inputs to the gRSC.

Keywords: Retrosplenial cortex, Value-based decision making, Hippocampal-cortical interaction

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Neural Mechanisms Linking Temporal and Directional Predictions in Sensory-Motor Behaviors

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Accurately predicting environmental changes is vital for our survival, especially when we need to rapidly adapt our actions based on incoming motion information. While previous research has examined predictions for individual motion components such as timing, direction, or speed, the common origin of these predictions remains uncertain. In this study, we explored the relationship between temporal and directional expectations in motion to determine if they share a common source. We conducted a visually-guided oculomotor behavioral experiment, manipulating expectations for motion timing and direction, while recording smooth pursuit eye movements and electroencephalography (EEG) data from thirty participants. Our findings indicate that temporal and directional expectations are linked by a common mechanism: a temporal expectation function derived from behavioral reaction time data could effectively explain directional expectation. We assessed the association between the two expectations across participants using a Bayesian model, revealing varying degrees of association. Through multivariate analysis of EEG data, we aimed to confirm the relationship between the sharpening of the population direction tuning function—underlying directional expectation—and the strength of temporal expectation. Our results showed that these factors covaried, with the degree of covariation depending on individual differences in the association between the two expectations, as estimated from behavioral data. Additionally, we observed that the strength of neural synchronization between the delta phase (related to direction tuning) and beta amplitude also varied with individual differences in the degree of association of expectations. Overall, our results suggest that different expectations may originate from a shared source and that beta oscillations could play a crucial role in this process.

Keywords: Expectation, Bayesian observer model, Sensorimotor processing

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The Double-Edged Sword of Event Segmentation in Children's Memory

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Children, like adults, naturally segment continuous events into discrete chunks in storing their experiences into memory. However, what defines an event boundary and how different types of boundaries interact is still under debate. We investigated the impact of story changes and spatial context changes on children's event memory. Some videos with story changes consisted of unrelated series of events while others had meaningful causal sequences (e.g., familiar event schemas). 89 children aged 4 to 9 viewed 1-min cartoon videos, each containing 3 event chunks marked by independent boundary raters. After every two videos, participants freely recalled what they had seen and completed temporal order judgment tasks of across-event and within-event scenes, as well as a scene recognition task with target scenes from the videos. We found that children remembered the temporal order better for the events with story-related boundaries but there was an interaction effect that spatial boundaries facilitated order memory for the videos with unrelated event sequences. We found similar effects in scene recognition as well. These findings together indicate that while children are guided by coherently connected storylines for episodic memory, spatial boundaries aid in event segmentation in the absence of such information. Moreover, for videos with unrelated event chunks, children remembered across-event order better than within-event order, particularly earlier in development. Recognition of target scenes for the targets was better than lures when the events contained spatial boundaries and unrelated event series. Both findings suggest that story changes or spatial context changes help children chunk their memories but may blur the resolution within each episodic chunk. By unveiling the effects of multiple event boundaries on children's memory, this study sheds light on the fundamental nature of the temporal structure of episodic memory from early in human development.

Keywords: Episodic memory development, Event segmentation, Spatial boundary

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Neurobehavioral impacts of a social observer on risky decision-making in cigarette smokers

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Smoking is a long-standing social problem that persists. Given that typical individuals adapt their behaviors to social environment and follow social norm, individuals' choices to smoke may originate from their social maladaptiveness. To examine smokers' (mal)adaptive use of social information in making their choices, we used a gambling task where individuals made choices under social observation. Never- and ever- smokers chose between a risky and a safe gamble. Before the task, participants were given an opportunity to learn the risk preferences of two partners—one risk-averse and one risk-seeking. On some trials during the gambling task, they were instructed that one of the two partners would observe their choices, while on other trials, no observer was present. Both groups identified which partner was risk-averse or seeking, but showed that their beliefs about the partners' true preferences were suboptimal. Using computational modeling, we showed that under social observation, never-smokers primarily relied on the partner's identity, i.e., more credible information than the partners' preferences. In contrast, ever-smokers were influenced by the partner's choice simulated based on the imperfect belief about their preferences. Event-related potential (ERP) patterns corroborated our behavioral results. Specifically, in never-smokers, P300 and N400, the ERP components known to be associated with attention and semantic processing, showed different amplitudes between observed and not observed trials at the time the identity of the partner was revealed. In ever-smokers, the ERP patterns between observed and not observed trials were not observed until the specific gamble options—information essential for simulating others' choices—were revealed. These results suggest that never-smokers adaptively used more precise and available social information in adjusting their choices, whereas ever-smokers relied on their imprecise belief about the social situation

Keywords: Smoking, Social context, EEG

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Enhanced Performance of Predictive Models in Within and Between-Person Pain using Multi-echo fMRI

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The predictive modeling approach in fMRI holds promise in identifying the relationship between brain and psychological states or behavior. However, inherent technical challenges in fMRI, particularly physiological and acquisition artifacts, lead to significant barriers to establishing a stable relationship between brain representations and corresponding psychological states and behaviors, necessitating more samples. Here, we employ multi-echo fMRI (ME-fMRI) combined with multi-echo Independent Component Analysis (ME-ICA) to overcome these challenges in developing predictive models. We developed two pattern-based fMRI models to predict within-person and between-person differences in pain. Our results indicate that ME-fMRI with ME-ICA outperforms conventional single-echo fMRI (SE-fMRI) in the predictive performances of both within-person and between-person pain models. We attribute these improvements to increased BOLD sensitivity, reflected in higher temporal signal-to-noise ratio, and reduced influence of individual-specific non-neuronal-like signals, such as head motion. Additionally, our findings suggest that ME-fMRI with ME-ICA can achieve comparable predictive performance to SE-fMRI with fewer trials and samples, highlighting its cost efficiency. Overall, combining ME-fMRI with ME-ICA shows promise in capturing both intra- and inter-individual variability in psychological states and behaviors.

Keywords: predictive modeling, pain, multi-echo fMRI

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Dynamic Pain Regulation: From Onset to Recovery

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Pain perception encompasses the entire process from onset to recovery. Previous studies on pain neuroimaging have mainly focused on brain systems during pain perception. However, due to the short inter-stimulus intervals (ITI) typically used in pain experiments, enough time for the brain to recover from the pain was not guaranteed. To investigate the entire pain-to-recover processes in the brain, we used four different levels of heat pain stimulus with a long ITI of 60 seconds (n=70). Participants continuously report their intensity rating from the anticipation period to the end of the trial. We employed the independent component analysis (ICA) method to define axes that explain pain processing and examined which sub-processes they account for. Based on the pre-defined axis, we aimed to construct task-related subspaces that systematically capture individual recovery trajectories. Overall, this study provides new insight into our understanding of the brain's dynamic recovery process, especially in a systematic view. These findings can ultimately lead to new approaches to pain treatment and recovery.

Keywords: pain regulation, dynamical system, ICA

Acknowledgements:

Neural Mechanisms of Music-Induced Analgesic Effect through Personally Pleasant Music

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Pleasant music is known to be effective for pain treatment. Recent research has shown that listening to music releases dopamine in the brain's reward system, suggesting the role of music's rewarding properties in the pain-relieving effect. However, the underlying neural mechanisms of how music effectively reduces pain remain largely unexplored. In this study, we aim to elucidate the neural mechanisms of music-induced analgesia. In the fMRI scanner, participants received thermal pain and capsaicin while listening to self-selected pleasant music and reported their perceived pain intensity. In addition to the pleasant music, the scrambled versions of those music were also provided as a control condition. With the data (current $n = 39$), first, we investigate whether there are significant differences in pain ratings between conditions. Next, we compare the expression of the Neural Pain Signature (NPS) and the Tonic Pain Signature (ToPS) to see if they are reduced with self-selected music compared to scrambled music. Then, we examine the mechanisms by which self-selected music effectively reduces pain, focusing on the involvement of dopamine-related brain regions. Finally, we explore what specific factors in music contribute to pain relief, distinguishing them from other elements such as placebo effects or self-regulation. This study can improve our understanding of the mechanisms by which music can relieve pain and elucidate its therapeutic potential.

Keywords: Music, Pain, fMRI

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Context-dependent influence of CRH neurons in the paraventricular nucleus of the hypothalamus on active avoidance behavior

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The paraventricular nucleus of the hypothalamus (PVN) plays a crucial role in the stress response as part of the hypothalamic-pituitary-adrenal (HPA) axis. Specifically, corticotropin-releasing hormone (CRH) neurons in the PVN (CRHPVN) not only stimulate the release of stress hormones, but also allow the appropriate action in stress situations independent of hormonal actions. However, the specific impact of CRHPVN on active defensive avoidance remains unclear. To address this, we selectively inactivated synaptic transmission of CRHPVN neurons using the Cre-dependent expression of the tetanus toxin (TetTox) in CRH-IRES-Cre mice. Mice were subjected to an active-avoidance test in a footshock chamber with panels creating a safe zone. Training sessions involved the presentation of CS which was 15-s tone stimuli paired with co-terminating US which was 10-s footshock (0.3mA). Active avoidance behaviour was quantified by measuring the number of times the mice fled to the safe zone during CS presentations. Following 3 days of training, the TetTox group showed significantly reduced active avoidance and increased freezing behaviour compared to the control group, indicating that silencing CRHPVN neurons impaired active defensive behaviour. The findings highlight that CRHPVN does exert a certain influence on active defensive behaviour.

Keywords: Paraventricular nucleus of the hypothalamus, corticotropin-releasing hormone neurons, Active avoidance.

Acknowledgements:

Neural Signatures of Individual Differences in the Influence of Interoceptive Signals on Preference Decisions

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Interoception, the ability to sense visceral signals, plays a pivotal role in recognizing oneself as an integrated individual, working in conjunction with exteroception. Previous research has shown that individual differences in interoception influence various psychological functions, including preference decisions. This study aims to identify the neural mechanisms involved in individual differences in the influence of interoceptive signals on preference and perceptual decisions. Participants were instructed to choose between two movie titles based on either subjective preferences or the luminance of the characters written during fMRI scanning. In each task condition, half of the trials were the congruent condition. The loss of efficiency in incongruent compared to the congruent condition during luminance task was used to assess the interference of stimulus-related external and self-related internal information. Interoceptive sensitivity was measured through the heartbeat counting task and self-reported survey. Repeated-measures analysis of variance on inverse efficiency scores revealed a significant congruency effect (CE) in both task conditions, indicating that incongruency between preference and luminance information led to a loss of decision-making efficiency. Individuals with lower CE in the luminance task showed higher interoceptive awareness in both subjective and objective measures. Conjunction and contrast analysis of neural features at the preference and luminance decision onsets revealed that individuals with larger CE in the luminance task tend to involve the Superior Parietal Lobule, post- and precentral gyrus during both preference and luminance decisions. In contrast, individuals with lower CE in the luminance task showed higher activation at the rostromedial PFC region during preference decisions compared to perceptual decisions.

Keywords: Interoceptive sensitivity, Preference, rmPFC

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Effects of Ketamine treatment on the uncontrollable stress-induced memory impairments

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Recently, Ketamine has attracted considerable interest for its therapeutic potential in treating depression and various mental health conditions. Single-dose treatment of Ketamine, one of the N-methyl-D-aspartate receptor (NMDAR) antagonists, has demonstrated rapid and sustained effects for depression, unlike traditional antidepressants. Stress is a prominent cause of depression, and stressed animals have been used to reveal brain mechanisms mediating depression and its therapeutic development. Therefore, we investigated the effects of Ketamine treatment on stress-induced cognitive impairment. Rats administered Ketamine intravenously (10mg/kg, i.v) were subjected to a 60-minute restraint plus 60 intermittent tail shocks (60-min stress), resulting in memory impairment. The cognitive status of these stressed rats was assessed using a novel object recognition task. Control rats (saline; i.v) showed impaired recognition memory following 60-min stress. However, rats with ketamine treatment showed no recognition memory impairments. Our findings suggest that intravenous ketamine treatments increased stress resistance.

Keywords: Ketamine, Stress, NOR

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Neural representations in MPFC and insula encode individual differences in estimating others' preferences

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In human society, successful social interactions hinge on appropriate reactions achieved through accurate estimation of other's perspectives, often necessitating the consideration of contextual information. The study delves into the neural mechanism underpinning this cognitive capacity, employing a preference estimation task. In this task, participants, upon seeing the target's face, are tasked with predicting the extent to which the target would prefer a given item. Preference estimation accuracy is quantified by the percentage of correct guesses, aligning with the target's preferences on a 4-point Likert scale.

The investigation employs both univariate and multivariate analyses. Univariate findings reveal the involvement of subregions of the medial prefrontal cortex (mPFC) in precisely assessing others' preferences. Multivariate results, utilizing inter-subject representational similarity analysis (IS-RSA), demonstrate that the multi-voxel patterns in the pregenual anterior cingulate cortex (pgACC) and the anterior insula (AI) predict individual variability in preference estimation accuracy. This implies that the diverse behavioral patterns among participants in inferring others' preferences were mirrored in the multivariate neural representations in these areas, both of which are strongly associated with individual differences in interoception and context-dependent ambiguous facial emotion estimation.

The present study suggests that the mPFC and AI play a pivotal role in accurately estimating others' preferences with minimal information. The application of multivariate pattern analysis unveils insights beyond the scope of traditional approaches, shedding light on the intricate neural processes involved in this social cognitive task.

Keywords: Anterior Cingulate Cortex, Insula, Interoception

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Effect of physical effort on dopamine activities in the ventral and dorsal striatums

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How do we learn to invest our time and effort? Reinforcement learning (RL) theory explains how the animals or agents can update the value through a teaching signal, reward prediction error (RPE). Separate lines of research show that effort exertion discounts the value of future reward, while retrospectively adding subjective value to it. Since evidence shows that slowly fluctuating striatal dopamine signal during approach to reward is consistent with RPE, it is no surprise that dopamine decreases as effort discounts the reward value. However, whether striatal dopamine accounts for effort is unclear.

To address this issue, we recorded dopamine response from the ventral striatum (VS) or dorsomedial striatum (DMS) in 18 mice performing behavioral tasks. While previous studies have used lever-pressing or T-maze tasks, we trained head-fixed mice for approaching a goal location to obtain a reward in virtual reality (VR). VR setup enables us to employ continuous visual stimuli as well as continuous effort cost by applying friction to the treadmill. As a result, we could observe the effect of effort cost on the slowly fluctuating dopamine activity while animals approach the goal location.

Firstly, we applied a constant level of friction (on or off) in a blockwise manner. Ramping dopamine signals in both VS and DMS decreased when friction was applied. Increasing levels of friction lead to systematic decrease in VS dopamine ramping signal up until the reward delivery. DMS dopamine response showed a similar pattern around the start position. However, the responses reached a similar level briefly before reward delivery. When visual cues were associated with a larger reward size, overall ramping response increased in both VS and DMS. Our results suggest that DMS dopamine activity may reflect physical effort cost, while VS dopamine reliably encodes reward prediction error.

Keywords: Effort, RPE, Dopamine

Acknowledgements:

A derivative-like dopaminergic computation across valences

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It is essential for an individual's survival to evaluate the surrounding situations and behave adaptively based on learned values. Previous studies have shown that mid-brain dopaminergic neuronal activity can provide a moment-by-moment teaching signal for value updating in rewarding situations. Such teaching signal is known as a temporal difference (TD) reward prediction error (RPE), which is roughly a derivative of a value function. Here, we examined whether dopamine activity in a situation with negative valence can also be explained by TD errors. We designed an active avoidance task in a virtual linear track with a beginning, shock, and safe zone. Mice had to run out of the shock zone to avoid electric tail shocks (0.8mA, 150ms, 3.5s interval). During task performance, dopaminergic activity in the ventral striatum was measured with dopamine sensor (DA2m) using fiber photometry. We observed gradual inhibition followed by excitation ($n = 10$ mice), consistent with TD error predictions. Manipulating the visual scene speed (slow, standard, fast) amplified both early inhibitory and later excitatory dopamine responses, aligning with the notion that value function slope increases with speed. Teleporting mice forward or backward caused phasic excitation and inhibition ($n = 5$ mice), reflecting sudden value changes. These results confirm the idea that dopamine activity during aversive situations follows a moment-by-moment derivative of the value function. Averaged neural data was fitted through model fit analysis to predict the signal by defining a value function. Recovered value function showed a sink-down shape in the middle, as predicted. Finally, we added a second track where mice received a reward 1.5s after the shock zone exit. We observed increased dopamine response before reward compared to the conditions without reward ($n = 4$ mice). This suggests that ventral striatum dopamine activity integrates both positive and negative valences via a unified derivative process.

Keywords: RPE, Dopamine, Avoidance, Ventral Striatum

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From vision to action during closed-loop naturalistic interaction: Spatially specific reward signals in visual areas and their behavioral impact

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The human visual system actively integrates information from the environment to guide goal-directed actions, with reward information playing a critical role in linking sensory inputs to optimal behaviors. While animal studies have demonstrated visual areas' sensitivity to rewards, the mechanisms by which potential rewards modulate visual representations during dynamic, naturalistic goal-directed actions in humans remain elusive. To address this gap, we developed a naturalistic 3D interactive paradigm as part of the 7T Naturalistic Perception, Action, and Cognition (NatPAC) dataset. In this study, participants completed a "shepherding" task in a Minecraft-based environment, herding sheep (rewards) to a goal location while avoiding risks (puddles and a fox), as we collected human fMRI and eye-tracking data. Combining high-precision gaze data with population receptive field (pRF) mapping, we predicted BOLD signals based on gaze-centered visual inputs. By subtracting these predictions from actual BOLD signals, we removed retinotopic visual stimulation effects and modeled the residuals using reward-related regressors. To examine spatial specificity of rewards, we segmented the visual field into eight radial bins and explained their variance with reward-related regressors. Results revealed significant spatial specificity for reward loss in visual areas, including V1, V2, and V3, as well as in the IPS, and FEF. Crucially, higher reward-based spatial specificity in these regions correlated with more successful loss-avoidant behaviors. Our findings demonstrate that visual areas encode spatially specific reward signals, which facilitate reward-maximizing behaviors in dynamic environments. This study highlights the intricate interplay between vision, action, and cognition in naturalistic settings, an aspect rarely explored in conventional laboratory paradigms.

Keywords: Naturalistic, Closed-loop, fMRI

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Value representation in striatum-projecting PPC neurons

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Estimation of action values requires the accumulation of the outcome history based on the previous choices. Previous studies in rats, monkeys, and humans have found the action value signals encoded in multiple regions of the brain, and these brain structures bias animals' choices toward more highly expected rewards. Recently, it has been shown that the neurons in the posterior parietal cortex (PPC) that project to the dorsomedial striatum (DMS) (PPC- $\rightarrow$ DMS) encode the history of the previous choices and influence current choices during a perceptual decision-making task. However, it is unclear whether the PPC- $\rightarrow$ DMS neurons encode the action values. To examine this, we developed a two-alternative forced choice (2AFC) task using a T-maze system, which can measure the left-right choices of freely navigating mice based on the reward probabilities that change across the session. We then measured the calcium activities of the PPC- $\rightarrow$ DMS neurons in task-performing mice. Interestingly, the PPC- $\rightarrow$ DMS neurons showed strong choice selectivity before or after the reversal of the reward probabilities, but the selectivity became weaker during value changes. We further quantified the action values of the task-performing mice using a model-free reinforcement learning algorithm and analyzed whether the activity of the PPC- $\rightarrow$ DMS neurons represents the action values using a generalized linear model. Our data showed that the PPC- $\rightarrow$ DMS neurons represent action values that are critical for animals' choices during the task. Our results suggest that the PPC- $\rightarrow$ DMS circuit plays a key role in value processing and making decisions in conditions when the reward probability changes.

Keywords: decision making, value processing, history, cortico-striatal projection, posterior parietal cortex, dorsomedial striatum

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The role of CGRP neurons in suppressing addictive behavior

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Behavioral modification such as the curbing of addictive behavior can be induced through electric shock or physical punishment by activating peripheral nociceptors, but this can cause unintended pain. Our study investigated a neural circuit that can suppress addictive behavior while minimizing nociceptive pain. Recent research has shown that calcitonin gene-related peptide (CGRP) in the parabrachial nucleus (PBN) generates general alarm signals and relays these signals to the forebrain, particularly targeting regions such as the central amygdala. Here, we investigated whether stimulating CGRP neurons can suppress addictive behavior without eliciting nocifensive reactions, using two approaches: 1) optogenetic stimulation and 2) cocaine self-administration. Through Calca-Cre::DAT-Cre mice injected with channelrhodopsin-2 (ChR2) in the VTA and PBN, we performed instrumental conditioning. Addictive lever-pressing behavior induced by unilateral optical stimulation of midbrain dopamine cells (1s, 20Hz) was significantly reduced by bilateral optical stimulation of CGRP neurons (3s, 30Hz). Furthermore, we observed that behavior resulting from self-administration of the actual drug cocaine (0.2 mg/kg/infusion) was also effectively suppressed. In addition, these behavioral modification effects had a long-lasting impact, persisting even weeks after the final stimulation session. Taken together, our findings demonstrate that activation of CGRP neurons can effectively suppress addictive behavior without triggering nocifensive responses.

Keywords: Addiction, Suppression, CGRP

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Modulation of inequity aversion through social hierarchy in mouse model

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Inequity aversion refers to the emotional, cognitive, and behavioral response to unfair outcomes. Several studies have argued that social hierarchy exerts a powerful influence on behavior. However, the relationship between social hierarchy and social inequity aversion, as well as the underlying neural mechanisms, remain unclear. We aim to determine the social hierarchy, investigate preferences for social inequity according to hierarchy, and confirm the corresponding brain activity. In this study, we used the tube test to determine the social hierarchy status of mice and then evaluated the mice's preference for the equitable versus inequitable chambers. The results indicated that subordinate mice showed a higher preference for the equitable condition compared to dominant mice, while dominant mice did not exhibit a clear preference for either the equitable or inequitable condition. This suggests that mice lower in social hierarchy are more sensitive to equity. When investigating brain activity via immunohistochemistry, we found that expression of c-Fos in the medial prefrontal cortex (mPFC), striatum, nucleus accumbens (NAc) were higher in subordinate mice compared to dominant ones. In summary, this study demonstrates that responses to inequity aversion can be modulated by social hierarchy, thus providing neurophysiological insight into the cognitive processes related to inequity. These findings offer an important foundation for models of social context perception based on social hierarchy.

Keywords: Cortex, Inequity aversion, Social hierarchy

Acknowledgements:

Shared memory representations in real-world social interactions

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Social interactions, particularly conversations, shape shared memories among individuals. These interactions are influenced by participants' diverse social traits and intentions, contributing to shared memories through participation in the same social events or similar scenarios. This study investigates the abstraction level in shared memory representations among participants who interacted together and those engaged in similar social scenarios with different individuals, modeling their interaction dynamics. Over three days, participants engaged in interactive social tasks in groups of three. On the first day, real-world social interactions within each group were recorded by head-mounted cameras. On the second day, participants recounted their experiences and viewed first-person perspective video replays in an fMRI scanner. The third day involved participants segmenting events from their video transcripts and recalls. To explore shared event representations, a language model, SBERT, was used to generate semantic embeddings for events from replays and recalls, comparing similarities for three types of event pairs: 1) recall-replay pairs within the same participants, 2) recall events within the same group, and 3) recall events across different groups. Results revealed higher neural pattern similarities in the posterior medial cortex for recall-replay pairs with high content similarity, indicating shared event representations. These findings suggest that abstract event representations are shared among individuals experiencing analogous social events, transcending specific episodic details. Furthermore, we analyzed group decision-making dynamics, demonstrating that participants' traits contributed to individual behaviors underlying group interactions. Using a social RL model, we replicated the observed social dynamics, providing a novel framework for understanding the complexity of conversation-based social interactions and their potential impact on shared memories.

Keywords: social interactions, memory, fMRI

Acknowledgements:

Dysfunction of the primary visual cortex leads to visual impairments in a Grin2b-mutant mouse model of autism

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Atypical sensory processing is a part of the diagnostic criteria for autism spectrum disorders (ASD), and deficits in sensory processing are common symptoms found in patients with ASD. However, the neural basis underlying sensory abnormalities in ASD is not clearly understood. One of the well-known ASD risk genes is Grin2b, which encodes the GluN2B subunit of the glutamate ionotropic NMDA receptors. In this study, we examined abnormalities of Grin2b-mutant mice having patient-derived C456Y mutation (Grin2bC456Y/+) during visual processing and discrimination behaviors. The mutant mice exhibited delayed learning and reduced performance in a visual discrimination task, even though their eye movement in the optokinetic response (OKR) was unaltered. Using in vivo two-photon calcium imaging, we found that the neurons in the primary visual cortex (V1) of the Grin2b-mutant mice showed an increased number of visual-responsive neurons but reduced stimulus-evoked activation and prolonged inhibition. Overall orientation selectivity of V1 neurons in Grin2b-mutant mice was normal. Collectively, our findings suggest that Grin2b-mutant mice exhibit reduced E/I balance during visual processing in the V1, which potentially leads to delayed perceptual learning.

Keywords: Autism, Visual cortex, two-photon calcium imaging

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Suppression of Risky Foraging Behavior Requires Increased Central Amygdala Activity at Discrete Time Points

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The ability of animals to inhibit ongoing behavior when facing a threat is crucial for survival. The central nucleus of the amygdala (CeA) is involved in regulating responses to both positive and negative stimuli. Nonetheless, the detailed mechanisms by which CeA activity inhibits ongoing behavior in response to threats remain under-explored. To address this, we utilized a paradigm where a predatory attack is simulated by a robot followed by the licking of a sucrose solution. Our results demonstrate that the robotic threat decreases rats' approach and consumption behaviors. This suppression of foraging behavior is attenuated when the CeA is electrically lesioned. Through fiber photometry, we identify CeA neurons that preferentially respond during reward consumption and threat presentation, with CeA activity further increasing in trials where the threat is presented after reward consumption. This heightened CeA activity is observed only when animals successfully avoid the threat prior to the robot's attack. Optogenetic inhibition of the CeA at discrete time points reveals that blocking CeA activity within the first 3 seconds of licking behavior before the robot attack disrupts behavior suppression, underscoring the importance of CeA activity at specific moments for suppressing behavior in response to a threat.

Keywords: Risky foraging, Amygdala, Predator like robot

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Weakening of cortical connectivity and multisensory integration in the Shank3 Δ 14–16 mice

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Dysfunction in multisensory integration is often observed in patients with autism spectrum disorder (ASD). Deletion in the Shank3 gene is well-known to cause autistic phenotypes in both humans and mice. However, it is still unclear whether and how Shank3 deficiency causes changes in anatomical and functional connectivity in the cortical areas that integrate multisensory information. One of the key areas that are important for multisensory integration is the posterior parietal cortex (PPC). In this study, we examined the dysfunction in multisensory behaviors and cortical projections to the PPC throughout the brain in the Shank3 deleted (Shank3 Δ 14–16) mice. We first found that Shank3 deletion in mice disrupted multisensory integration-dependent behaviors. Next, we anatomically traced the neurons that project to the PPC across cortical areas and found that the cortical inputs to the PPC were significantly reduced in the Shank3 Δ 14–16 mice. Finally, we used wide-field calcium imaging and measured functional connectivity between the PPC and other cortical regions. We found a significant reduction in the calcium activity correlation between the frontal motor cortex and the PPC in Shank3 Δ 14–16 mice. Collectively, our data suggest that weakened anatomical and functional connectivity in the PPC may cause multisensory dysfunctions in Shank3 Δ 14–16 mice.

Keywords: Autism spectrum disorder, Multisensory integration, Posterior parietal cortex

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Plcg1 dependent neural plasticity is required for dynamic hippocampal CA2 network processing social memory in mice

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A growing body of evidence suggests that the hippocampal CA2 region is crucial for processing social memory, including discerning novel social stimuli from familiar ones (retrieving social memory) and becoming familiarized with novel social stimuli (acquiring social memory). Previous studies have shown that the representations of CA2 neural networks differ when presented with familiar versus novel social stimuli. However, the precise moment when a novel stimulus becomes familiar has not been explored. Here, we investigated how the CA2 neural network changes during the acquisition of social stimuli. To achieve this, we imaged the hippocampal CA2 calcium activity in mice in vivo during repetitive presentations of a novel conspecific until it became familiar. We found that the configuration of the hippocampal CA2 neural ensemble changes as the mice become familiarized with the presented conspecific. Interestingly, transgenic mice with Plcg1 knockdown in hippocampal CA2, previously shown to be unable to acquire social memory, did not show changes in the hippocampal CA2 neural ensemble during this repeated exposure to the same conspecific. These results suggest novel mechanisms for how social information is processed in the hippocampal CA2 region.

Keywords: Hippocampus, social memory, in vivo Calcium imaging

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Chemogenetic activation of the lateral habenula facilitates the association of CS with the absence of the US in Pavlovian appetitive conditioning

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The presentations of an unconditioned stimulus (US) contingent upon the occurrence of a conditioned stimulus (CS) establishes excitatory conditioning or conditioned excitation (e.g., approaching a food-cup area). Conversely, the association of a CS with the nonoccurrence of a US generates inhibitory conditioning or conditioned inhibition (e.g., not approaching a food-cup area). We have demonstrated that experimental treatments impairing lateral habenula (LHb) function interfered with the establishment of conditioned inhibition. Therefore, the present study investigated the effects of chemogenetic LHb activation on the establishment of conditioned inhibition. The experimental design consisted of two phases: Animals first received the explicitly unpaired training (light-nonoccurrence of food) for three days along with chemogenetic LHb activation. This was followed by six days of paired training (light-occurrence of food). Animals for genetic control did not exhibit the retardation-of-acquisition in the pairing learning, whereas animals for the chemogenetic LHb activation did. These results indicated that chemogenetic LHb activation during three days of unpaired training delayed the acquisition of subsequent excitatory conditioning. These findings suggest that LHb activation during unpaired training facilitates the formation of conditioned inhibition. These findings contribute to our understanding of the neural mechanisms underlying conditioned inhibition and shed light on the role of the LHb in this process.

Keywords: Lateral Habenula, DREADDs, Conditioned inhibition

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Egocentric spatial information decoding by population activity in the mouse retrosplenial cortex

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Within the retrosplenial cortex neuronal populations, various types of spatially tuned neurons exist including egocentric boundary vector cells, egocentric border cells, and egocentric vertex cells and head direction cells, among others. However, the collective role of these diverse spatially tuned neurons in encoding egocentric spatial information of environmental geometry in a two-dimensional space remains unclear. To address this question, we analyzed the population activity recorded from GCaMP-expressing RSC excitatory neuron while mice freely navigated in open chamber. Using Bayesian decoding of population activity, the egocentric spatial information of boundaries and vertices of the environmental geometry could be successfully decoded. This result suggests that that spatially tuned RSC neurons may work together to encoding the environmental geometry in the egocentric reference frame.

Keywords: Retrosplenial Cortex, CALCIUM IMAGING

Acknowledgements:

Roles of retrosplenial parvalbumin-positive interneurons in the neural representation of environmental geometry

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Effective navigation in complex environments relies on the construction of an egocentric cognitive map. This map requires the integration of information on environmental geometry, such as vertices and boundaries, in an egocentric reference frame. Recent studies reported that subpopulations of excitatory neurons in the retrosplenial cortex (RSC) are selectively activated near specific distances and orientations relative to boundaries and vertices of environmental geometry, encoding the geometry in an egocentric manner. However, the roles of GABAergic interneurons in egocentric spatial navigation are yet unknown. In this study, we performed in vivo Ca²⁺ imaging of GCaMP-expressing parvalbumin-positive (PV) interneurons in freely-moving PV-Cre mice during the exploration of a square open chamber. Additionally, optogenetic inactivation of PV was conducted to elucidate how PV interneurons affect the spatial information of excitatory neurons by recording excitatory neurons during PV inactivation. We found that subpopulations of PV interneurons selectively activated near geometric boundaries and vertices in an egocentric manner, similar to excitatory neurons. Furthermore, optogenetic inactivation of PV changed the spatial receptive fields of excitatory neurons in either a broader or shifting in the opposite direction. Overall, these results indicate PV interneurons in the RSC can encode spatial information egocentrically and significantly influence the spatial coding of excitatory neurons.

Keywords: Egocentric Vertex Cell ,Retrosplenial Cortex , Interneuron

Acknowledgements:

Impaired Egocentric Neural Encoding of Environmental Geometry in the Retrosplenial Cortex in the 5XFAD Mouse Model of Alzheimer's Disease

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Egocentric spatial encoding neurons have been found in the retrosplenial cortex (RSC), which play a crucial role in the formation of an egocentric cognitive map. Notably, the RSC is among the initial brain regions impacted by amyloid beta accumulation in Alzheimer's disease (AD), which is marked by impairments in episodic memory and spatial navigation. However, the effect of amyloidosis in AD on egocentric neural encoding remains unclear. To investigate this, we conducted in vivo Ca²⁺ imaging of GCaMP-expressing excitatory neurons in the RSC as both control mice and 5XFAD mice freely explored a square open chamber. Our preliminary findings suggest that egocentric neural encoding of environmental geometry is compromised in 5XFAD mice compared to control mice, suggesting that amyloidosis disrupts egocentric spatial representation.

Keywords: Egocentric Vertex Cell , Retrosplenial Cortex , Alzheimer's disease

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